Literature search report – run 20260828T160426

scitech-librarian 3.2

August 28, 2026

Generated by scitech-librarian 3.2 (report level: intermediate). Every number below is reproducible from the archived run directory `20260828T160426`.

Item	Value
Run started	2026-08-28 16:04:26
Duration	42 s
Query file	queries.example.json
Blocks	PSC, MLP, NOV, QC
Backends / sources	openalex, arxiv, inspire
Mode	full fetch, up to 300 records per block and backend
Non-curated venue filter	on
Open-access lookup	not run
Journal metrics	on file for 103 journals
Filters	none
Interrupted	no

Search strategy

Each block is one structural query – a conjunction of synonym groups, (a OR b) AND (c OR d) – rendered into every backend's native grammar. The strings below are exactly what was sent (PRISMA-S item 8).

Block PSC: Perovskite solar cell degradation under humidity

Purpose: grounding block – expect thousands of hits; use it to sanity-check that every backend is reachable and the counts are plausible

```
(perovskite solar cell OR halide perovskite photovoltaic) AND (degradation OR stability OR encapsulation)
 AND (humidity OR moisture OR water ingress)
```

Backend	Query string sent
openalex	("perovskite solar cell" OR "halide perovskite photovoltaic") AND (degradation OR stability OR encapsulation) AND (humidity OR moisture OR "water ingress")
arxiv	(all:"perovskite solar cell" OR all:"halide perovskite photovoltaic") AND (all:"degradation" OR all:"stability" OR all:"encapsulation")
inspire	("perovskite solar cell" or "halide perovskite photovoltaic") and ("degradation" or "stability" or "encapsulation") and ("humidity" or "moisture" or "water ingress")

Block MLP: Machine-learned interatomic potentials for amorphous materials

Purpose: a narrower intersection – expect hundreds; good for testing record fetch and RIS export

```
(machine learning potential OR machine-learned interatomic potential OR neural network potential) AND
(amorphous OR glass OR disordered solid) AND (molecular dynamics OR structure prediction)
```

arXiv receives groups [0, 1] only (nested-boolean limitation).

Backend	Query string sent
openalex	("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND (amorphous OR glass OR "disordered solid") AND ("molecular dynamics" OR "structure prediction")
arxiv	(all:"machine learning potential" OR all:"machine-learned interatomic potential" OR all:"neural network potential") AND (all:"amorphous" OR all:"glass" OR all:"disordered solid")
inspire	("machine learning potential" or "machine-learned interatomic potential" or "neural network potential") and ("amorphous" or "glass" or "disordered solid") and ("molecular dynamics" or "structure prediction")

Block NOV: EXAMPLE NOVELTY CHECK: origami metamaterials as topological acoustic pumps

Purpose: novelty-check pattern – a SMALL number is the good outcome; read every hit by hand before claiming a gap

```
(origami OR kirigami) AND (acoustic metamaterial OR phononic crystal) AND (topological pumping OR
Thouless pump OR edge state)
```

Backend	Query string sent
openalex	(origami OR kirigami) AND ("acoustic metamaterial" OR "phononic crystal") AND ("topological pumping" OR "Thouless pump" OR "edge state")
arxiv	(all:"origami" OR all:"kirigami") AND (all:"acoustic metamaterial" OR all:"phononic crystal")
inspire	("origami" or "kirigami") and ("acoustic metamaterial" or "phononic crystal") and ("topological pumping" or "Thouless pump" or "edge state")

Block QC: Quasicrystal photonics (arXiv-tuned example)

Purpose: demonstrates arxiv_groups: arXiv chokes on deeply nested booleans, so name the two groups it should get

```
(quasicrystal OR quasiperiodic lattice) AND (photonic OR waveguide OR optical lattice) AND (localization
OR critical states OR fractal spectrum)
```

arXiv receives groups [0, 2] only (nested-boolean limitation).

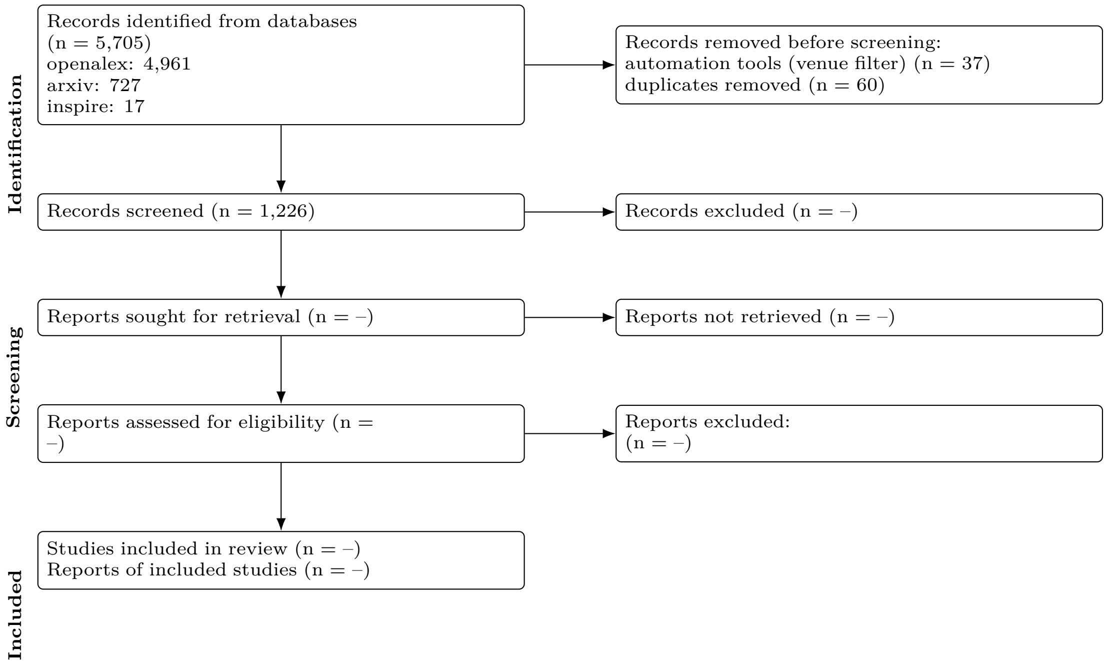
Backend	Query string sent
openalex	(quasicrystal OR "quasiperiodic lattice") AND (photonic OR waveguide OR "optical lattice") AND (localization OR "critical states" OR "fractal spectrum")
arxiv	(all:"quasicrystal" OR all:"quasiperiodic lattice") AND (all:"localization" OR all:"critical states" OR all:"fractal spectrum")
inspire	("quasicrystal" or "quasiperiodic lattice") and ("photonic" or "waveguide" or "optical lattice") and ("localization" or "critical states" or "fractal spectrum")

Results summary

Block	openalex	arxiv	inspire	Identified	Retrieved	Unique
PSC	4569	175	0	4,744	475	474
MLP	238	138	0	376	344	313
NOV	0	0	0	0	0	0
QC	154	414	17	585	467	439
Total	4,961	727	17	5,705	1286	1226

Identified = database hit counts (not comparable across backends: proximity operators are dropped and stemming differs). Retrieved = records actually downloaded after the venue filter, capped by `--limit`. Unique = after DOI/title deduplication across all sources.

PRISMA 2020 flow



Stage	n
Records identified from databases	5,705
openalex	4,961
arxiv	727
inspire	17
Records retrieved (downloaded / ingested)	1,323
Removed before screening: automation (non-curated venues)	37
Removed before screening: duplicates	60
Records to screen (unique)	1,226
Records screened	1,226 (assumed = unique)
Records excluded at screening	–
Reports sought for retrieval	–
Reports not retrieved	–
Reports assessed for eligibility	–
Studies included	–
Reports of included studies	–

Automation stages are computed from the data; '–' marks manual stages not yet recorded in prisma.json. Note that 'identified' counts hits reported by each database while 'retrieved' is what was downloaded within `--limit`, so the two differ on large blocks.

PRISMA-S search-reporting checklist

Item	Requirement	This search
1	Database name	openalex, arxiv, inspire (documented public APIs)
2	Multi-database searching	3 databases, one structural query per block rendered into each native grammar; see Search strategy
3	Study registries	not applicable
4	Online resources and browsing	none recorded
5	Citation searching	not performed
6	Contacts	none recorded
7	Other methods	none
8	Full search strategies	reported verbatim per backend under Search strategy; archived in queries.json
9	Limits and restrictions	record download capped at 300 per block and backend, most-cited first; no date, language or document-type limits applied
10	Search filters	venue filter on: records from non-curated repositories (Zenodo, Figshare, SSRN...) removed
11	Prior work	none
12	Updates	3 earlier run(s) archived; counts tracked in counts_history.csv
13	Dates of searches	searched on 2026-08-28 16:04:26
14	Peer review	none
15	Total records	5,705 identified from databases; 1,323 retrieved; 1,226 unique
16	Deduplication	exact DOI match, else first 90 characters of the lower-cased title; 60 duplicates removed

Records

Deduplicated across sources, sorted by cited.

Block PSC (474 unique)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Incorporation of rubidium cations into perovskite solar cells improves photovoltaic performance	Michael Saliba; Taisuke Matsui; Konrad Domanski et al.	2016	Science	3691	10.1126/science.aah5557	
High-efficiency two-dimensional Ruddlesden–Popper perovskite solar cells	Hsinhan Tsai; Wanyi Nie; Jean-Christophe Blancon et al.	2016	Nature	3361	10.1038/nature18306	
Efficient, stable and scalable perovskite solar cells using poly(3-hexylthiophene)	Eui Hyuk Jung; Nam Joong Jeon; Eun Young Park et al.	2019	Nature	2318	10.1038/s41586-019-1036-3	
A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability	Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.	2014	Angewandte Chemie International Edition	2006	10.1002/anie.201406466	[illegible] (2026)
Review of recent progress in chemical stability of perovskite solar cells	Guangda Niu; Xudong Guo; Liduo Wang	2014	Journal of Materials Chemistry A	1945	10.1039/c4ta04994b	
Understanding Degradation Mechanisms and Improving Stability of Perovskite Photovoltaics	Caleb C. Boyd; Rongrong Cheacharoen; Tomas Leijtens et al.	2018	Chemical Reviews	1767	10.1021/acs.chemrev.8b00336	53.8[illegible] (2026)
Formamidinium and Cesium Hybridization for Photo- and Moisture-Stable Perovskite Solar Cell	Jin-Wook Lee; Deok-Hwan Kim; Hui-Seon Kim et al.	2015	Advanced Energy Materials	1661	10.1002/aenm.201501310	[illegible] (2026)
23.6%-efficient monolithic perovskite/silicon tandem solar cells with improved stability	Kevin A. Bush; Axel F. Palmstrom; Zhengshan J. Yu et al.	2017	Nature Energy	1533	10.1038/nenergy.2017.9	
Investigation of CH 3 NH 3 PbI 3 Degradation Rates and Mechanisms in Controlled Humidity Environments Using in Situ Techniques	Jinli Yang; Braden D. Siempelkamp; Dianyi Liu et al.	2015	ACS Nano	1342	10.1021/nn506864k	16.894 (2026)
Carbon Nanotube/Polymer Composites as a Highly Stable Hole Collection Layer in Perovskite Solar Cells	Severin N. Habisreutinger; Tomas Leijtens; Giles E. Eperon et al.	2014	Nano Letters	1198	10.1021/nl501982b	
Improved performance and stability of perovskite solar cells by crystal crosslinking with alkylphosphonic acid -ammonium chlorides	Xiong Li; M. Ibrahim Dar; Chenyi Yi et al.	2015	Nature Chemistry	1176	10.1038/nchem.2324	
All-Inorganic Perovskite Solar Cells	Jia Liang; Caixing Wang; Yanrong Wang et al.	2016	Journal of the American Chemical Society	1092	10.1021/jacs.6b10227	
Study on the stability of CH 3 NH 3 PbI 3 films and the effect of post-modification by aluminum oxide in all-solid-state hybrid solar cells	Guangda Niu; Wenzhe Li; Fanqi Meng et al.	2013	Journal of Materials Chemistry A	1067	10.1039/c3ta13606j	
Organometallic-functionalized interfaces for highly efficient inverted perovskite solar cells	Zhen Li; Bo Li; Xin Wu et al.	2022	Science	1031	10.1126/science.abm8566	
Degradation observations of encapsulated planar CH 3 NH 3 PbI 3 perovskite solar cells at high temperatures and humidity	Yu Han; S. Meyer; Yasmina Dkhissi et al.	2015	Journal of Materials Chemistry A	997	10.1039/c5ta00358j	
Stability of perovskite solar cells	Dian Wang; Matthew Wright; Naveen Kumar Elumalai et al.	2016	Solar Energy Materials and Solar Cells	993	10.1016/j.solmat.2015.12.025	
Strained hybrid perovskite thin films and their impact on the intrinsic stability of perovskite solar cells	Jingjing Zhao; Yehao Deng; Haotong Wei et al.	2017	Science Advances	965	10.1126/sciadv.aao5616	
Scalable fabrication and coating methods for perovskite solar cells and solar modules	Nam-Gyu Park; Kai Zhu	2020	Nature Reviews Materials	902	10.1038/s41578-019-0176-2	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Improving efficiency and stability of perovskite solar cells with photocurable fluoropolymers	Federico Bella; Gianmarco Griffini; Juan-Pablo Correa-Baena et al.	2016	Science	866	10.1126/science.aah4046	
Thermochromic halide perovskite solar cells	Jia Lin; Minliang Lai; Letian Dou et al.	2018	Nature Materials	854	10.1038/s41563-017-0006-0	
Damp heat–stable perovskite solar cells with tailored-dimensionality 2D/3D heterojunctions	Randi Azmi; Esma Ugur; Akmaral Seitkhan et al.	2022	Science	843	10.1126/science.abm5784	
Tailored interfaces of unencapsulated perovskite solar cells for >1,000 hour operational stability	Jeffrey A. Christians; Philip Schulz; Jonathan S. Tinkham et al.	2018	Nature Energy	826	10.1038/s41560-017-0067-y	
Stable High-Performance Perovskite Solar Cells via Grain Boundary Passivation	Tianqi Niu; Jing Lü; Rahim Munir et al.	2018	Advanced Materials	800	10.1002/adma.2017[illegible]	[illegible] (2026)
In Situ Growth of 2D Perovskite Capping Layer for Stable and Efficient Perovskite Solar Cells	Peng Chen; Yang Bai; Songcan Wang et al.	2018	Advanced Functional Materials	772	10.1002/adfm.2017[illegible]	[illegible] (2026)
2D perovskite stabilized phase-pure formamidinium perovskite solar cells	Jin-Wook Lee; Zhenghong Dai; Tae-Hee Han et al.	2018	Nature Communications	747	10.1038/s41467-018-05454-4	
Suppression of atomic vacancies via incorporation of isovalent small ions to increase the stability of halide perovskite solar cells in ambient air	Makhsud I. Saidaminov; Junghwan Kim; Ankit Jain et al.	2018	Nature Energy	745	10.1038/s41560-018-0192-2	
A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability	Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.	2014	Angewandte Chemie	692	10.1002/ange.2014[illegible]	[illegible] (2026)
Stable high efficiency two-dimensional perovskite solar cells via cesium doping	Xu Zhang; Xiaodong Ren; Bin Liu et al.	2017	Energy & Environmental Science	675	10.1039/c7ee0114[illegible]	[illegible]0.475 (2026)
A polymer scaffold for self-healing perovskite solar cells	Yicheng Zhao; Jing Wei; Heng Li et al.	2016	Nature Communications	666	10.1038/ncomms10228	
Accelerated degradation of methylammonium lead iodide perovskites induced by exposure to iodine vapour	Shenghao Wang; Yan Jiang; Emilio J. Juárez-Pérez et al.	2016	Nature Energy	657	10.1038/nenergy.2016.195	
All-Inorganic CsPbX 3 Perovskite Solar Cells: Progress and Prospects	Jingru Zhang; Gary Hodes; Zhiwen Jin et al.	2019	Angewandte Chemie International Edition	611	10.1002/anie.2019[illegible]	[illegible] (2026)
Intact 2D/3D halide junction perovskite solar cells via solid-phase in-plane growth	Yeoun-Woo Jang; Seungmin Lee; Kyung Mun Yeom et al.	2021	Nature Energy	606	10.1038/s41560-020-00749-7	
Trapped charge-driven degradation of perovskite solar cells	Namyoung Ahn; Kwisung Kwak; Min Seok Jang et al.	2016	Nature Communications	601	10.1038/ncomms13422	
Stability of Perovskite Solar Cells: A Prospective on the Substitution of the A Cation and X Anion	Ze Wang; Zejiao Shi; Taotao Li et al.	2016	Angewandte Chemie International Edition	595	10.1002/anie.2016[illegible]	[illegible] (2026)
Efficient and stable perovskite solar cells prepared in ambient air irrespective of the humidity	Qidong Tai; Peng You; Hongqian Sang et al.	2016	Nature Communications	582	10.1038/ncomms11105	
Efficient and stable Ruddlesden–Popper perovskite solar cell with tailored interlayer molecular interaction	Hui Ren; Shidong Yu; Lingfeng Chao et al.	2020	Nature Photonics	581	10.1038/s41566-019-0572-6	
Quantum-size-tuned heterostructures enable efficient and stable inverted perovskite solar cells	Hao Chen; Sam Teale; Bin Chen et al.	2022	Nature Photonics	581	10.1038/s41566-022-00985-1	
Perovskite-polymer composite cross-linker approach for highly-stable and efficient perovskite solar cells	Tae-Hee Han; Jin-Wook Lee; Chungseok Choi et al.	2019	Nature Communications	581	10.1038/s41467-019-08455-z	
Stabilizing the Efficiency Beyond 20% with a Mixed Cation Perovskite Solar Cell Fabricated in Ambient Air under Controlled Humidity	Trilok Singh; Tsutomu Miyasaka	2017	Advanced Energy Materials	573	10.1002/aenm.2017[illegible]	[illegible] (2026)
Two-dimensional Ruddlesden–Popper layered perovskite solar cells based on phase-pure thin films	Chao Liang; Hao Gu; Yingdong Xia et al.	2020	Nature Energy	554	10.1038/s41560-020-00721-5	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Low-loss contacts on textured substrates for inverted perovskite solar cells	So Min Park; Mingyang Wei; Nikolaos Lempesis et al.	2023	Nature	548	10.1038/s41586-023-06745-7	
Multifunctional Chemical Linker Imidazoleacetic Acid Hydrochloride for 21% Efficient and Stable Planar Perovskite Solar Cells	Jiangzhao Chen; Xing Zhao; Seul-Gi Kim et al.	2019	Advanced Materials	534	10.1002/adma.20192290862 (2026)	
A review on perovskite solar cells: Evolution of architecture, fabrication techniques, commercialization issues and status	Priyanka Roy; Numeshwar Kumar Sinha; Sanjay Tiwari et al.	2020	Solar Energy	530	10.1016/j.solener.2020.01.080	
CsPb0.9Sn0.1IBr2 Based All-Inorganic Perovskite Solar Cells with Exceptional Efficiency and Stability	Jia Liang; Peiyang Zhao; Caixing Wang et al.	2017	Journal of the American Chemical Society	528	10.1021/jacs.7b07949	
Functionalization of perovskite thin films with moisture-tolerant molecules	Shuang Yang; Yun Wang; Porun Liu et al.	2016	Nature Energy	527	10.1038/nenergy.2015.16	
Surface Passivation Using 2D Perovskites toward Efficient and Stable Perovskite Solar Cells	Guangbao Wu; Rui Liang; Mingzheng Ge et al.	2021	Advanced Materials	524	10.1002/adma.2022105635 (2026)	
Engineering Stress in Perovskite Solar Cells to Improve Stability	Nicholas Rolston; Kevin A. Bush; Adam D. Printz et al.	2018	Advanced Energy Materials	510	10.1002/aenm.2018802139(2026)	
Interfacial Residual Stress Relaxation in Perovskite Solar Cells with Improved Stability	Hao Wang; Cheng Zhu; Lang Liu et al.	2019	Advanced Materials	496	10.1002/adma.20192.4068 (2026)	
Enhanced Efficiency and Stability of Inverted Perovskite Solar Cells Using Highly Crystalline SnO 2 Nanocrystals as the Robust Electron-Transporting Layer	Zonglong Zhu; Yang Bai; Xiao Liu et al.	2016	Advanced Materials	496	10.1002/adma.20162.0819 (2026)	
Degradation mechanism of hybrid tin-based perovskite solar cells and the critical role of tin (IV) iodide	Luis Lanzetta; Thomas Webb; Nourdine Zibouche et al.	2021	Nature Communications	493	10.1038/s41467-021-22864-z	
Precise Control of Crystal Growth for Highly Efficient CsPbI2Br Perovskite Solar Cells	Weijie Chen; Haiyang Chen; Guiying Xu et al.	2018	Joule	491	10.1016/j.joule.2018.10.011	
Gas chromatography–mass spectrometry analyses of encapsulated stable perovskite solar cells	Lei Shi; Martin P. Bucknall; Trevor L. Young et al.	2020	Science	489	10.1126/science.aba2412	
Engineering ligand reactivity enables high-temperature operation of stable perovskite solar cells	So Min Park; Mingyang Wei; Jian Xu et al.	2023	Science	482	10.1126/science.adi4107	
Advances in hole transport materials engineering for stable and efficient perovskite solar cells	Zinab H. Bakr; Qamar Wali; Azhar Fakharuddin et al.	2017	Nano Energy	448	10.1016/j.nanoen.2017.02.025	
Flexible Perovskite Solar Cells	Hyun Suk Jung; Gill Sang Han; Nam-Gyu Park et al.	2019	Joule	447	10.1016/j.joule.2019.07.023	
Perovskite Solar Cell Stability in Humid Air: Partially Reversible Phase Transitions in the PbI 2 -CH 3 NH 3 I-H 2 O System	Zhaoning Song; Antonio Abate; Suneth C. Watthage et al.	2016	Advanced Energy Materials	435	10.1002/aenm.20160846(2026)	
Is Excess PbI2 Beneficial for Perovskite Solar Cell Performance?	Fangzhou Liu; Qi Dong; Man Kwong Wong et al.	2016	Advanced Energy Materials	422	10.1002/aenm.201582206(2026)	
Device stability of perovskite solar cells – A review	Muhammad Imran Asghar; Jun Zhang; H. Wang et al.	2017	Renewable and Sustainable Energy Reviews	418	10.1016/j.rser.2017.04.003	
Multifunctional Enhancement for Highly Stable and Efficient Perovskite Solar Cells	Yuan Cai; Jian Cui; Ming Chen et al.	2020	Advanced Functional Materials	418	10.1002/adfm.202005976 (2026)	
Enhancing stability and efficiency of perovskite solar cells with crosslinkable silane-functionalized and doped fullerene	Yang Bai; Qingfeng Dong; Yuchuan Shao et al.	2016	Nature Communications	416	10.1038/ncomms12806	
Tailored Amphiphilic Molecular Mitigators for Stable Perovskite Solar Cells with 23.5% Efficiency	Hongwei Zhu; Yuhang Liu; Felix T. Eickemeyer et al.	2020	Advanced Materials	416	10.1002/adma.20190757 (2026)	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Unveiling facet-dependent degradation and facet engineering for stable perovskite solar cells	Chunqing Ma; Felix T. Eickemeyer; Sun-Ho Lee et al.	2023	Science	415	10.1126/science.adf3349	
2D/3D heterojunction engineering at the buried interface towards high-performance inverted methylammonium-free perovskite solar cells	Jing Li; Cong Zhang; Cheng Gong et al.	2023	Nature Energy	395	10.1038/s41560-023-01295-8	
Orientation Regulation of Phenylethylammonium Cation Based 2D Perovskite Solar Cell with Efficiency Higher Than 11%	Xinqian Zhang; Gang Wu; Weifei Fu et al.	2018	Advanced Energy Materials	389	10.1002/aenm.201[illegible]	[illegible] (2026)
Isomer-Pure Bis-PCBM-Assisted Crystal Engineering of Perovskite Solar Cells Showing Excellent Efficiency and Stability	Fei Zhang; Wenda Shi; Jingshan Luo et al.	2017	Advanced Materials	388	10.1002/adma.201[illegible]	[illegible] (2026)
UV Degradation and Recovery of Perovskite Solar Cells	Sang-Won Lee; Seongtak Kim; Soohyun Bae et al.	2016	Scientific Reports	383	10.1038/srep38150	
Polymer Doping for High-Efficiency Perovskite Solar Cells with Improved Moisture Stability	Jiexuan Jiang; Qian Wang; Zhiwen Jin et al.	2017	Advanced Energy Materials	378	10.1002/aenm.201[illegible]	[illegible] (2026)
A chemically inert bismuth interlayer enhances long-term stability of inverted perovskite solar cells	Shaohang Wu; Rui Chen; Shasha Zhang et al.	2019	Nature Communications	378	10.1038/s41467-019-09167-0	
Humidity-Induced Degradation via Grain Boundaries of HC(NH2)2PbI3 Planar Perovskite Solar Cells	Jae Sung Yun; Jincheol Kim; Trevor L. Young et al.	2018	Advanced Functional Materials	373	10.1002/adfm.201[illegible]	[illegible] (2026)
Enhancing Stability of Perovskite Solar Cells to Moisture by the Facile Hydrophobic Passivation	Insung Hwang; Inyoung Jeong; Jinwoo Lee et al.	2015	ACS Applied Materials & Interfaces	353	10.1021/acsami.5b[illegible]	[illegible] (2026)
Optimal Interfacial Engineering with Different Length of Alkylammonium Halide for Efficient and Stable Perovskite Solar Cells	Hyeonwoo Kim; Seungun Lee; Do Yoon Lee et al.	2019	Advanced Energy Materials	353	10.1002/aenm.201[illegible]	[illegible] (2026)
Passivation of Grain Boundaries by Phenethylammonium in Formamidinium-Methylammonium Lead Halide Perovskite Solar Cells	Da Seul Lee; Jae Sung Yun; Jincheol Kim et al.	2018	ACS Energy Letters	349	10.1021/acsenergy[illegible]	[illegible] (2026)
Material and Device Stability in Perovskite Solar Cells	Hui-Seon Kim; Ja-Young Seo; Nam-Gyu Park	2016	ChemSusChem	347	10.1002/cssc.201[illegible]	[illegible] (2026)
Mixed Dimensional 2D/3D Hybrid Perovskite Absorbers: The Future of Perovskite Solar Cells?	Anurag Krishna; Sébastien Gottis; Mohammad Khaja Nazeeruddin et al.	2018	Advanced Functional Materials	347	10.1002/adfm.201[illegible]	[illegible] (2026)
Mixed Cation FAxPEA1–xPbI3 with Enhanced Phase and Ambient Stability toward High-Performance Perovskite Solar Cells	Nan Li; Zonglong Zhu; Chu-Chen Chueh et al.	2016	Advanced Energy Materials	345	10.1002/aenm.201[illegible]	[illegible] (2026)
Improvement of the humidity stability of organic–inorganic perovskite solar cells using ultrathin Al 2 O 3 layers prepared by atomic layer deposition	Xu Dong; Xiang Fang; Minghang Lv et al.	2015	Journal of Materials Chemistry A	343	10.1039/c4ta06128d	
Encapsulation for long-term stability enhancement of perovskite solar cells	Fabio Matteocci; Lucio Cinà; Enrico Lamanna et al.	2016	Nano Energy	337	10.1016/j.nanoen.2016.09.041	
Phase Pure 2D Perovskite for High-Performance 2D–3D Heterostructured Perovskite Solar Cells	Pengwei Li; Yiqiang Zhang; Chao Liang et al.	2018	Advanced Materials	336	10.1002/adma.201[illegible]	[illegible] (2026)
Boosting the efficiency of carbon-based planar CsPbBr3 perovskite solar cells by a modified multistep spin-coating technique and interface engineering	Xingyue Liu; Xianhua Tan; Zhiyong Liu et al.	2018	Nano Energy	336	10.1016/j.nanoen.2018.11.053	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
High-Efficiency Perovskite Solar Cells with Imidazolium-Based Ionic Liquid for Surface Passivation and Charge Transport	Xuejie Zhu; Minyong Du; Jiangshan Feng et al.	2020	Angewandte Chemie International Edition	335	10.1002/anie.2020159876 (2026)	
Mixed 3D–2D Passivation Treatment for Mixed-Cation Lead Mixed-Halide Perovskite Solar Cells for Higher Efficiency and Better Stability	Yongyoon Cho; Arman Mahboubi Soufiani; Jae Sung Yun et al.	2018	Advanced Energy Materials	334	10.1002/aenm.20178332 (2026)	
Role of 4- tert -Butylpyridine as a Hole Transport Layer Morphological Controller in Perovskite Solar Cells	Shen Wang; Mahsa Sina; Pritesh Parikh et al.	2016	Nano Letters	333	10.1021/acs.nanolett.6b02158	
Room-Temperature Molten Salt for Facile Fabrication of Efficient and Stable Perovskite Solar Cells in Ambient Air	Lingfeng Chao; Yingdong Xia; Bixin Li et al.	2019	Chem	324	10.1016/j.chempr.2019.02.025	
Design of low bandgap tin–lead halide perovskite solar cells to achieve thermal, atmospheric and operational stability	Rohit Prasanna; Tomas Leijtens; Sean P. Dunfield et al.	2019	Nature Energy	323	10.1038/s41560-019-0471-6	
Fundamental Understanding of Photocurrent Hysteresis in Perovskite Solar Cells	Pengyun Liu; Wei Wang; Shaomin Liu et al.	2019	Advanced Energy Materials	321	10.1002/aenm.20188317 (2026)	
Red-Carbon-Quantum-Dot-Doped SnO2 Composite with Enhanced Electron Mobility for Efficient and Stable Perovskite Solar Cells	Wei Hui; Yingguo Yang; Quan Xu et al.	2019	Advanced Materials	313	10.1002/adma.20190874 (2026)	
Reduction of bulk and surface defects in inverted methylammonium- and bromide-free formamidinium perovskite solar cells	Rui Chen; Jianan Wang; Zonghao Liu et al.	2023	Nature Energy	299	10.1038/s41560-023-01288-7	
High-Performance Perovskite Solar Cells with Enhanced Environmental Stability Based on a (p-FC6H4C2H4NH3)2[PbI4] Capping Layer	Qin Zhou; Lusheng Liang; Junjie Hu et al.	2019	Advanced Energy Materials	295	10.1002/aenm.20182595 (2026)	
Stable and Efficient Organo-Metal Halide Hybrid Perovskite Solar Cells via -Conjugated Lewis Base Polymer Induced Trap Passivation and Charge Extraction	Pingli Qin; Guang Yang; Zhiwei Ren et al.	2018	Advanced Materials	293	10.1002/adma.20172086 (2026)	
Unveiling the Role of tBP–LiTFSI Complexes in Perovskite Solar Cells	Shen Wang; Zihuan Huang; Xuefeng Wang et al.	2018	Journal of the American Chemical Society	290	10.1021/jacs.8b09809	
High-Performance Perovskite Solar Cells with Excellent Humidity and Thermo-Stability via Fluorinated Perylenediimide	Jia Yang; Cong Liu; C.S. Cai et al.	2019	Advanced Energy Materials	289	10.1002/aenm.20190398 (2026)	
Strain in perovskite solar cells: origins, impacts and regulation	Jinpeng Wu; Shunchang Liu; Zongbao Li et al.	2021	National Science Review	289	10.1093/nsr/nwab047	
Encapsulation of Organic and Perovskite Solar Cells: A Review	Ashraf Uddin; Mushfika Baishakhi Upama; Haimang Yi et al.	2019	Coatings	285	10.3390/coatings90065 (2026)	
Metal Oxide Charge Transport Layers for Efficient and Stable Perovskite Solar Cells	Seong Sik Shin; Seon Joo Lee; Sang Il Seok	2019	Advanced Functional Materials	285	10.1002/adfm.201900455 (2026)	
Encapsulation for improving the lifetime of flexible perovskite solar cells	Hasitha C. Weerasinghe; Yasmina Dkhissi; Andrew D. Scully et al.	2015	Nano Energy	284	10.1016/j.nanoen.2015.10.006	
Fluorinated Low-Dimensional Ruddlesden–Popper Perovskite Solar Cells with over 17% Power Conversion Efficiency and Improved Stability	Jishan Shi; Yerun Gao; Xiang Gao et al.	2019	Advanced Materials	283	10.1002/adma.20190873 (2026)	
Influence of Electrode Interfaces on the Stability of Perovskite Solar Cells: Reduced Degradation Using MoOx/Al for Hole Collection	Erin M. Sanehira; Bertrand J. Tremolet de Villers; Philip Schulz et al.	2016	ACS Energy Letters	282	10.1021/acsenergylett.6b00128 (2026)	

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Highly Air-Stable Carbon-Based -CsPbI3 Perovskite Solar Cells with a Broadened Optical Spectrum	Sisi Xiang; Zhongheng Fu; Weiping Li et al.	2018	ACS Energy Letters	279	10.1021/acsenergylett.8b00826 16.781 (2026)	
Fabrication of perovskite solar cells in ambient air by blocking perovskite hydration with guanabenz acetate salt	Luyao Yan; Hao Huang; Peng Cui et al.	2023	Nature Energy	278	10.1038/s41560-023-01358-w	
Interfacial Defect Passivation and Stress Release via Multi-Active-Site Ligand Anchoring Enables Efficient and Stable Methylammonium-Free Perovskite Solar Cells	Baibai Liu; Huan Bì; Dongmei He et al.	2021	ACS Energy Letters	276	10.1021/acsenergylett.1c00294 16.712 (2026)	
Suppressing Vacancy Defects and Grain Boundaries via Ostwald Ripening for High-Performance and Stable Perovskite Solar Cells	Yuqian Yang; Jihuai Wu; Xiaobing Wang et al.	2019	Advanced Materials	275	10.1002/adma.201904347 20.086 (2026)	
Quantum Dots Supply Bulk- and Surface-Passivation Agents for Efficient and Stable Perovskite Solar Cells	Xiaopeng Zheng; Joel Troughton; Nicola Gasparini et al.	2019	Joule	272	10.1016/j.joule.2019.05.005	
Bifunctional Organic Spacers for Formamidinium-Based Hybrid Dion–Jacobson Two-Dimensional Perovskite Solar Cells	Yang Li; Jovana V. Milić; Amita Ummadisingu et al.	2018	Nano Letters	271	10.1021/acs.nanolett.8b03552	
Over 20% PCE perovskite solar cells with superior stability achieved by novel and low-cost hole-transporting materials	Fei Zhang; Zhi-Qiang Wang; Hongwei Zhu et al.	2017	Nano Energy	269	10.1016/j.nanoen.2017.09.035	
Effect of Selective Contacts on the Thermal Stability of Perovskite Solar Cells	Xing Zhao; Hui-Seon Kim; Ja-Young Seo et al.	2017	ACS Applied Materials & Interfaces	264	10.1021/acsami.6b15673 8.387 (2026)	
Well-Defined Thiolated Nanographene as Hole-Transporting Material for Efficient and Stable Perovskite Solar Cells	Jing Cao; Yumin Liu; Xiaojing Jing et al.	2015	Journal of the American Chemical Society	261	10.1021/jacs.5b06493	
Stability Issues on Perovskite Solar Cells	Xing Zhao; Nam-Gyu Park	2015	Photonics	259	10.3390/photonics2041139	
Interface Modification by Ionic Liquid: A Promising Candidate for Indoor Light Harvesting and Stability Improvement of Planar Perovskite Solar Cells	Meng Li; Chao Zhao; Zhao-Kui Wang et al.	2018	Advanced Energy Materials	257	10.1002/aenm.201801509 18.63 (2026)	
Vertically Oriented 2D Layered Perovskite Solar Cells with Enhanced Efficiency and Good Stability	Xinqian Zhang; Gang Wu; Shida Yang et al.	2017	Small	255	10.1002/smll.201700611	
A Review: Thermal Stability of Methylammonium Lead Halide Based Perovskite Solar Cells	Tanzila Tasnim Ava; Abdullah Al Mamun; Sylvain Marsillac et al.	2019	Applied Sciences	254	10.3390/app9010188 3.811 (2026)	
Multifunctional Phosphorus-Containing Lewis Acid and Base Passivation Enabling Efficient and Moisture-Stable Perovskite Solar Cells	Zhi Yang; Jinjuan Dou; Song Kou et al.	2020	Advanced Functional Materials	254	10.1002/adfm.201910710 14.975 (2026)	
Highly Efficient and Stable Perovskite Solar Cells via Modification of Energy Levels at the Perovskite/Carbon Electrode Interface	Zhifang Wu; Zonghao Liu; Zhanhao Hu et al.	2019	Advanced Materials	254	10.1002/adma.201804284 20.086 (2026)	
Intermolecular Exchange Boosts Efficiency of Air-Stable, Carbon-Based All-Inorganic Planar CsPbIBr2 Perovskite Solar Cells to Over 9%	Weidong Zhu; Qianni Zhang; Dazheng Chen et al.	2018	Advanced Energy Materials	251	10.1002/aenm.201802080 18.63 (2026)	
Interfacial defect passivation and stress release by multifunctional KPF6 modification for planar perovskite solar cells with enhanced efficiency and stability	Huan Bì; Baibai Liu; Dongmei He et al.	2021	Chemical Engineering Journal	250	10.1016/j.cej.2021.129375 11.537 (2026)	

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Green-Solvent-Processable, Dopant-Free Hole-Transporting Materials for Robust and Efficient Perovskite Solar Cells	Junwoo Lee; Mahdi Malekshahi Byranvand; Gyeongho Kang et al.	2017	Journal of the American Chemical Society	250	10.1021/jacs.7b04949	
Investigation of Thermally Induced Degradation in CH3NH3PbI3 Perovskite Solar Cells using In-situ Synchrotron Radiation Analysis	Nam-Koo Kim; Young Hwan Min; Seokhwan Noh et al.	2017	Scientific Reports	249	10.1038/s41598-017-04690-w	
Inorganic Rubidium Cation as an Enhancer for Photovoltaic Performance and Moisture Stability of HC(NH2)2PbI3 Perovskite Solar Cells	Yun Hee Park; Inyoung Jeong; Seunghwan Bae et al.	2017	Advanced Functional Materials	248	10.1002/adfm.2016[illegible]	[illegible] (2026)
A Review on Encapsulation Technology from Organic Light Emitting Diodes to Organic and Perovskite Solar Cells	Qian Lu; Zhichun Yang; Xin Meng et al.	2021	Advanced Functional Materials	246	10.1002/adfm.202[illegible]	[illegible] (2026)
Accelerated Lifetime Testing of Organic–Inorganic Perovskite Solar Cells Encapsulated by Polyisobutylene	Lei Shi; Trevor L. Young; Jincheol Kim et al.	2017	ACS Applied Materials & Interfaces	244	10.1021/acsami.7b[illegible]	[illegible] (2026)
All-inorganic CsPbBr 3 perovskite solar cell with 10.26% efficiency by spectra engineering	Haiwen Yuan; Yuanyuan Zhao; Jialong Duan et al.	2018	Journal of Materials Chemistry A	242	10.1039/c8ta08900k	
Copper Salts Doped Spiro-OMeTAD for High-Performance Perovskite Solar Cells	Meng Li; Zhao-Kui Wang; Yingguo Yang et al.	2016	Advanced Energy Materials	241	10.1002/aenm.201[illegible]	[illegible] (2026)
A comprehensive device modelling of perovskite solar cell with inorganic copper iodide as hole transport material	Syed Zulqarnain Haider; Hafeez Anwar; Mingqing Wang	2018	Semiconductor Science and Technology	241	10.1088/1361-6641/aaa596	
Self-assembled bilayer for perovskite solar cells with improved tolerance against thermal stresses	Bitao Dong; Mingyang Wei; Yuheng Li et al.	2025	Nature Energy	240	10.1038/s41560-024-01689-2	
Boron Doping of Multiwalled Carbon Nanotubes Significantly Enhances Hole Extraction in Carbon-Based Perovskite Solar Cells	Xiaoli Zheng; Haining Chen; Qiang Li et al.	2017	Nano Letters	239	10.1021/acs.nanolett.7b00200	
FAPbI3-Based Perovskite Solar Cells Employing Hexyl-Based Ionic Liquid with an Efficiency Over 20% and Excellent Long-Term Stability	Seçkin Akın; Erdi Akman; Savaş Sönmezoğlu	2020	Advanced Functional Materials	236	10.1002/adfm.2020[illegible]	[illegible] (2026)
Enhancing Stability of Perovskite Solar Cells to Moisture by the Facile Hydrophobic Passivation	; ; et al.	2015		235	(link)	
Inorganic CsPb1−xSnxIBr2 for Efficient Wide-Bandgap Perovskite Solar Cells	Nan Li; Zonglong Zhu; Jiangwei Li et al.	2018	Advanced Energy Materials	235	10.1002/aenm.201[illegible]	[illegible] (2026)
Review on Chemical Stability of Lead Halide Perovskite Solar Cells	Jing Zhuang; Jizheng Wang; Feng Yan	2023	Nano-Micro Letters	232	10.1007/s40820-023-01046-0	
Enhancement of thermal stability for perovskite solar cells through cesium doping	Guangda Niu; Wenzhe Li; Jiangwei Li et al.	2017	RSC Advances	231	10.1039/c6ra28501e	
Inhibition of halide oxidation and deprotonation of organic cations with dimethylammonium formate for air-processed p–i–n perovskite solar cells	Hongguang Meng; Kaitian Mao; Fengchun Cai et al.	2024	Nature Energy	229	10.1038/s41560-024-01471-4	
Perovskite Solar Cells: A Review of the Recent Advances	Priyanka Roy; Aritra Ghosh; Fraser Barclay et al.	2022	Coatings	228	10.3390/coatings1[illegible]	[illegible] (2026)
Crown Ether Modulation Enables over 23% Efficient Formamidinium-Based Perovskite Solar Cells	Tzu-Sen Su; Felix T. Eickemeyer; Michael A. Hope et al.	2020	Journal of the American Chemical Society	227	10.1021/jacs.0c08592	

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Large-area perovskite solar cells employing spiro-Naph hole transport material	Mingyu Jeong; In Woo Choi; Kanghoon Yim et al.	2022	Nature Photonics	225	10.1038/s41566-021-00931-7	
Compositional Engineering for Thermally Stable, Highly Efficient Perovskite Solar Cells Exceeding 20% Power Conversion Efficiency with 85 °C/85% 1000 h Stability	Taisuke Matsui; Teruaki Yamamoto; Takashi Nishihara et al.	2019	Advanced Materials	222	10.1002/adma.20180682320.683 (2026)	
High-Performance Flexible Perovskite Solar Cells via Precise Control of Electron Transport Layer	Keqing Huang; Yongyi Peng; Yaxin Gao et al.	2019	Advanced Energy Materials	221	10.1002/aenm.20190141918.63 (2026)	
Encapsulation of Perovskite Solar Cells for High Humidity Conditions	Qi Dong; Fangzhou Liu; Man Kwong Wong et al.	2016	ChemSusChem	219	10.1002/cssc.2016004785.866 (2026)	
Perovskite Solar Cells: A Review of the Latest Advances in Materials, Fabrication Techniques, and Stability Enhancement Strategies	Rakesh A. Afre; Diego Pugliese	2024	Micromachines	218	10.3390/mi15020192	
A Low-Temperature Thin-Film Encapsulation for Enhanced Stability of a Highly Efficient Perovskite Solar Cell	Young Il Lee; Nam Joong Jeon; Bong Jun Kim et al.	2017	Advanced Energy Materials	217	10.1002/aenm.20170192818.63 (2026)	
Electric-Field-Induced Degradation of Methylammonium Lead Iodide Perovskite Solar Cells	Soohyun Bae; Seongtak Kim; Sang-Won Lee et al.	2016	The Journal of Physical Chemistry Letters	216	10.1021/acs.jpclett.6b01176	
Efficient and stable tin-based perovskite solar cells by introducing -conjugated Lewis base	Tianhao Wu; Xiao Liu; Xin He et al.	2019	Science China Chemistry	212	10.1007/s11426-019-9653-8	
Tailoring the Functionality of Organic Spacer Cations for Efficient and Stable Quasi-2D Perovskite Solar Cells	Weifei Fu; Hongbin Liu; Xueliang Shi et al.	2019	Advanced Functional Materials	211	10.1002/adfm.20190022714.975 (2026)	
Importance of Functional Groups in Cross-Linking Methoxysilane Additives for High-Efficiency and Stable Perovskite Solar Cells	Lin Xie; Jiangzhao Chen; Parth Vashishtha et al.	2019	ACS Energy Letters	211	10.1021/acsenergylett.9b0165616.72 (2026)	
Perovskite Solar Cells with Inorganic Electron- and Hole-Transport Layers Exhibiting Long-Term (500 h) Stability at 85 °C under Continuous 1 Sun Illumination in Ambient Air	Seongrok Seo; Seonghwa Jeong; Changdeuck Bae et al.	2018	Advanced Materials	209	10.1002/adma.20180108120.683 (2026)	
Thermodynamically Self-Healing 1D–3D Hybrid Perovskite Solar Cells	Jiandong Fan; Yunping Ma; Cuiling Zhang et al.	2018	Advanced Energy Materials	208	10.1002/aenm.20170363118.63 (2026)	
Hydrophobic Organic Hole Transporters for Improved Moisture Resistance in Metal Halide Perovskite Solar Cells	Tomas Leijtens; Tommaso Giovenzana; Severin N. Habisreutinger et al.	2016	ACS Applied Materials & Interfaces	207	10.1021/acsami.5b103898.823 (2026)	
Wafer-scale monolayer MoS 2 film integration for stable, efficient perovskite solar cells	Huachao Zai; Pengfei Yang; Jie Su et al.	2025	Science	207	10.1126/science.ado2351	
Synergistic Effects of Eu-MOF on Perovskite Solar Cells with Improved Stability	Jie Dou; Cheng Zhu; Hao Wang et al.	2021	Advanced Materials	206	10.1002/adma.20210294720.683 (2026)	
Molecular materials as interfacial layers and additives in perovskite solar cells	Maria Vasilopoulou; Azhar Fakharuddin; Athanassios G. Coutsolelos et al.	2020	Chemical Society Reviews	206	10.1039/c9cs0073342.455 (2026)	
Continuous Grain-Boundary Functionalization for High-Efficiency Perovskite Solar Cells with Exceptional Stability	Yingxia Zong; Yuanyuan Zhou; Yi Zhang et al.	2018	Chem	204	10.1016/j.chempr.2018.03.00511.8 (2026)	
In situ studies of the degradation mechanisms of perovskite solar cells	Soumya Kundu; Timothy L. Kelly	2020	EcoMat	201	10.1002/eom2.1202515.557 (2026)	

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Passivation of the grain boundaries of CH 3 NH 3 PbI 3 using carbon quantum dots for highly efficient perovskite solar cells with excellent environmental stability	Qiang Guo; Fanglong Yuan; Bing Zhang et al.	2018	Nanoscale	200	10.1039/c8nr08295b	
Synergistic Effect of Fluorinated Passivator and Hole Transport Dopant Enables Stable Perovskite Solar Cells with an Efficiency Near 24%	Hongwei Zhu; Yameng Ren; Linfeng Pan et al.	2021	Journal of the American Chemical Society	200	10.1021/jacs.0c12802	
Chiral-structured heterointerfaces enable durable perovskite solar cells	Tianwei Duan; Shuai You; Min Chen et al.	2024	Science	199	10.1126/science.ado5172	
Compositional and Interface Engineering of Organic-Inorganic Lead Halide Perovskite Solar Cells	Haizhou Lu; Anurag Krishna; Shaik M. Zakeeruddin et al.	2020	iScience	198	10.1016/j.isci.2020.101359	
Efficient Passivation with Lead Pyridine-2-Carboxylic for High-Performance and Stable Perovskite Solar Cells	Sheng Fu; Xiaodong Li; Li Wan et al.	2019	Advanced Energy Materials	198	10.1002/aenm.201901852	19.685 (2026)
Highly Air-Stable Tin-Based Perovskite Solar Cells through Grain-Surface Protection by Gallic Acid	Tianyue Wang; Qidong Tai; Xuyun Guo et al.	2020	ACS Energy Letters	194	10.1021/acsenergylett.0c00256	16.710 (2026)
Propylammonium Chloride Additive for Efficient and Stable FAPbI3 Perovskite Solar Cells	Yong Zhang; Yan Li; Lin Zhang et al.	2021	Advanced Energy Materials	192	10.1002/aenm.202100338	18.253 (2026)
Highly Efficient and Stable Perovskite Solar Cells by Interfacial Engineering Using Solution-Processed Polymer Layer	Feijiu Wang; Ai Shimazaki; Fengjiu Yang et al.	2017	The Journal of Physical Chemistry C	191	10.1021/acs.jpcc.6b12137	
Poly(4-Vinylpyridine)-Based Interfacial Passivation to Enhance Voltage and Moisture Stability of Lead Halide Perovskite Solar Cells	Bhumika Chaudhary; Ashish Kulkarni; Ajay Kumar Jena et al.	2017	ChemSusChem	190	10.1002/cssc.201700276	5.471 (2026)
Monoammonium Porphyrin for Blade-Coating Stable Large-Area Perovskite Solar Cells with >18% Efficiency	Congping Li; Jun Yin; Ruihao Chen et al.	2019	Journal of the American Chemical Society	190	10.1021/jacs.9b01305	
Perovskite solar cells based on spiro-OMeTAD stabilized with an alkylthiol additive	Xu Liu; Bolin Zheng; Lei Shi et al.	2022	Nature Photonics	189	10.1038/s41566-022-01111-x	
NbF 5 : A Novel -Phase Stabilizer for FA-Based Perovskite Solar Cells with High Efficiency	Shihao Yuan; Qian Fang; Shaomin Yang et al.	2019	Advanced Functional Materials	189	10.1002/adfm.201807850	14.975 (2026)
Enhancing the Performance of Inverted Perovskite Solar Cells via Grain Boundary Passivation with Carbon Quantum Dots	Yuhui Ma; Heyi Zhang; Yewei Zhang et al.	2018	ACS Applied Materials & Interfaces	187	10.1021/acsami.8b13827	8.867 (2026)
Defect-Engineering-Enabled High-Efficiency All-Inorganic Perovskite Solar Cells	Jia Liang; Xiao Han; Ji-Hui Yang et al.	2019	Advanced Materials	186	10.1002/adma.201903448	22.068 (2026)
Interface Engineering of Imidazolium Ionic Liquids toward Efficient and Stable CsPbBr 3 Perovskite Solar Cells	Wenyu Zhang; Xiaojie Liu; Benlin He et al.	2020	ACS Applied Materials & Interfaces	186	10.1021/acsami.9b20381	8.831 (2026)
Review of Novel Passivation Techniques for Efficient and Stable Perovskite Solar Cells	Jincheol Kim; Anita Ho-Baillie; Shujuan Huang	2019	Solar RRL	185	10.1002/solr.201800302	
Thermal Stability of CuSCN Hole Conductor-Based Perovskite Solar Cells	Minsu Jung; Young Chan Kim; Nam Joong Jeon et al.	2016	ChemSusChem	183	10.1002/cssc.201600957	5.476 (2026)

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Effect of Side-Group-Regulated Dipolar Passivating Molecules on CsPbBr 3 Perovskite Solar Cells	Jialong Duan; Meng Wang; Yingli Wang et al.	2021	ACS Energy Letters	183	10.1021/acsenergylett.1c0[illegible]	[illegible] (2026)
Crystallization Kinetics in 2D Perovskite Solar Cells	Youkui Xu; Meng Wang; Yutian Lei et al.	2020	Advanced Energy Materials	181	10.1002/aenm.2020[illegible]	[illegible] (2026)
Efficiency and stability enhancement of perovskite solar cells by introducing CsPbI3 quantum dots as an interface engineering layer	Chang Liu; Manman Hu; Xianyong Zhou et al.	2018	NPG Asia Materials	181	10.1038/s41427-018-0055-0	
Water-Soluble Triazolium Ionic-Liquid-Induced Surface Self-Assembly to Enhance the Stability and Efficiency of Perovskite Solar Cells	Shuangjie Wang; Zhen Li; Yuanyuan Zhang et al.	2019	Advanced Functional Materials	181	10.1002/adfm.2019[illegible]	[illegible] (2026)
Perovskite solar cells: Materials, configurations and stability	Isabel Mesquita; Luísa Andrade; Adélio Mendes	2017	Renewable and Sustainable Energy Reviews	178	10.1016/j.rser.2017.09.011	
Enhancing Moisture and Water Resistance in Perovskite Solar Cells by Encapsulation with Ultrathin Plasma Polymers	Jesús Idígoras; Francisco J. Aparicio; Lidia Contreras-Bernal et al.	2018	ACS Applied Materials & Interfaces	177	10.1021/acsami.7b[illegible]	[illegible] (2026)
Enhanced Stability of Perovskite Solar Cells with Low-Temperature Hydrothermally Grown SnO2 Electron Transport Layers	Qin Liu; Minchao Qin; Weijun Ke et al.	2016	Advanced Functional Materials	176	10.1002/adfm.2016[illegible]	[illegible] (2026)
Targeted Therapy for Interfacial Engineering Toward Stable and Efficient Perovskite Solar Cells	Shuhui Wang; Haiyang Chen; Jiandong Zhang et al.	2019	Advanced Materials	176	10.1002/adma.2019[illegible]	[illegible] (2026)
Sulfonate-Assisted Surface Iodide Management for High-Performance Perovskite Solar Cells and Modules	Ruihao Chen; Yongke Wang; Siqing Nie et al.	2021	Journal of the American Chemical Society	176	10.1021/jacs.1c03419	
Recent efficient strategies for improving the moisture stability of perovskite solar cells	Faming Li; Mingzhen Liu	2017	Journal of Materials Chemistry A	169	10.1039/c7ta01325f	
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Spontaneously Self-Assembly of a 2D/3D Heterostructure Enhances the Efficiency and Stability in Printed Perovskite Solar Cells	Jinlong Hu; Chuan Wang; Shudi Qiu et al.	2020	Advanced Energy Materials	169	10.1002/aenm.2020[illegible]	[illegible] (2026)
Constructing Efficient and Stable Perovskite Solar Cells via Interconnecting Perovskite Grains	Xian Hou; Sumei Huang; Wei Ou-Yang et al.	2017	ACS Applied Materials & Interfaces	165	10.1021/acsami.7b[illegible]	[illegible] (2026)
Fluorine-Containing Passivation Layer via Surface Chelation for Inorganic Perovskite Solar Cells	Hao Zhang; Wanchun Xiang; Xuejiao Zuo et al.	2022	Angewandte Chemie International Edition	165	10.1002/anie.20221[illegible]	[illegible] (2026)
Reducing Energy Disorder in Perovskite Solar Cells by Chelation	Yiting Jiang; Jiabin Wang; Huachao Zai et al.	2022	Journal of the American Chemical Society	164	10.1021/jacs.1c12732	
Poly(N,N -bis-4-butylphenyl-N,N-bisphenyl)benzidine-Based Interfacial Passivation Strategy Promoting Efficiency and Operational Stability of Perovskite Solar Cells in Regular Architecture	N. N. Hisham; Sibel Akın-Basar [illegible]	2020	Advanced Materials	163	10.1002/adma.2020[illegible]	[illegible] (2026)
Enhanced Performance and Stability of 3D/2D Tin Perovskite Solar Cells Fabricated with a Sequential Solution Deposition	Efat Jokar; Po-Yuan Cheng; Chia-Yi Lin et al.	2021	ACS Energy Letters	161	10.1021/acsenergylett.1c0[illegible]	[illegible] (2026)
The Role of Charge Selective Contacts in Perovskite Solar Cell Stability	Bart Roose; Qiong Wang; Antonio Abate	2018	Advanced Energy Materials	155	10.1002/aenm.2018[illegible]	[illegible] (2026)
Over 24% efficient MA-free CsxFA1−xPbX3 perovskite solar cells	Siyang Wang; Liguo Tan; Junjie Zhou et al.	2022	Joule	155	10.1016/j.joule.2022.05.002	

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Self-Encapsulating Thermostable and Air-Resilient Semitransparent Perovskite Solar Cells	Jie Zhao; Kai Oliver Brinkmann; Ting Hu et al.	2017	Advanced Energy Materials	151	10.1002/aenm.2016[illegible]	[illegible] (2026)
Paradoxical Approach with a Hydrophilic Passivation Layer for Moisture-Stable, 23% Efficient Perovskite Solar Cells	Chunqing Ma; Nam-Gyu Park	2020	ACS Energy Letters	151	10.1021/acsenergylett.0c0[illegible]	16.7[illegible] (2026)
Research Update: Strategies for improving the stability of perovskite solar cells	Severin N. Habisreutinger; David P. McMeekin; Henry J. Snaith et al.	2016	APL Materials	150	10.1063/1.496121[illegible]	[illegible].589 (2026)
Dion-Jacobson 2D-3D perovskite solar cells with improved efficiency and stability	Xiaoqing Jiang; Jiafeng Zhang; Sajjad Ahmad et al.	2020	Nano Energy	150	10.1016/j.nanoen.2020.104892	
Perfluorinated Self-Assembled Monolayers Enhance the Stability and Efficiency of Inverted Perovskite Solar Cells	Christian M. Wolff; Laura Canil; Carolin Rehermann et al.	2020	ACS Nano	150	10.1021/acsnano.9b[illegible]	[illegible] (2026)
Improving Efficiency and Stability of Perovskite Solar Cells Enabled by A Near-Infrared-Absorbing Moisture Barrier	Qin Hu; Wei Chen; Wenqiang Yang et al.	2020	Joule	149	10.1016/j.joule.2020.06.007	
Review on persistent challenges of perovskite solar cells' stability	Maithili K. Rao; D. Sangeetha; M. Selvakumar et al.	2021	Solar Energy	149	10.1016/j.solener.2021.03.005	
High-Performance Thickness Insensitive Perovskite Solar Cells with Enhanced Moisture Stability	Jiehuan Chen; Lijian Zuo; Yingzhu Zhang et al.	2018	Advanced Energy Materials	148	10.1002/aenm.2018[illegible]	[illegible] (2026)
Moisture-Resistant FAPbI3 Perovskite Solar Cell with 22.25 % Power Conversion Efficiency through Pentafluorobenzyl Phosphonic Acid Passivation	Erdi Akman; Ahmed Esmail Shalan; Faranak Sadegh et al.	2020	ChemSusChem	148	10.1002/cssc.20200[illegible]	[illegible] (2026)
In situ growth of graphene on both sides of a Cu–Ni alloy electrode for perovskite solar cells with improved stability	Xuesong Lin; Hongzhen Su; Sifan He et al.	2022	Nature Energy	148	10.1038/s41560-022-01038-1	
Self-Seeding Growth for Perovskite Solar Cells with Enhanced Stability	Fei Zhang; Chuanxiao Xiao; Xihan Chen et al.	2019	Joule	147	10.1016/j.joule.2019.03.023	
Ionic liquids engineering for high-efficiency and stable perovskite solar cells	Xiaoyu Deng; Lisha Xie; Shurong Wang et al.	2020	Chemical Engineering Journal	146	10.1016/j.cej.2020.[illegible]	[illegible] (2026)
Multifunctional Small Molecule as Buried Interface Passivator for Efficient Planar Perovskite Solar Cells	Meizi Wu; Yuwei Duan; Lu Yang et al.	2023	Advanced Functional Materials	146	10.1002/adfm.2023[illegible]	[illegible] (2026)
1000 h Operational Lifetime Perovskite Solar Cells by Ambient Melting Encapsulation	Sai Ma; Yang Bai; Hao Wang et al.	2020	Advanced Energy Materials	145	10.1002/aenm.2019[illegible]	[illegible] (2026)
The balance between efficiency, stability and environmental impacts in perovskite solar cells: a review	Antonio Urbina	2019	Journal of Physics Energy	145	10.1088/2515-7655/ab5eee	
Dual-Protection Strategy for High-Efficiency and Stable CsPbI2Br Inorganic Perovskite Solar Cells	Sheng Fu; Wenxiao Zhang; Xiaodong Li et al.	2020	ACS Energy Letters	143	10.1021/acsenergylett.9b0[illegible]	16.7[illegible] (2026)
Quantifying the Interface Defect for the Stability Origin of Perovskite Solar Cells	Jionghua Wu; Jiangjian Shi; Yiming Li et al.	2019	Advanced Energy Materials	142	10.1002/aenm.2019[illegible]	[illegible] (2026)
Critical Role of Water in Defect Aggregation and Chemical Degradation of Perovskite Solar Cells	Yun-Hyok Kye; Chol-Jun Yu; Un-Gi Jong et al.	2018	The Journal of Physical Chemistry Letters	141	10.1021/acs.jpclett.8b00406	
Impact of Peripheral Groups on Phenothiazine-Based Hole-Transporting Materials for Perovskite Solar Cells	Fei Zhang; Shirong Wang; Hongwei Zhu et al.	2018	ACS Energy Letters	141	10.1021/acsenergylett.8b00[illegible]	16.7[illegible] (2026)
Dually-Passivated Perovskite Solar Cells with Reduced Voltage Loss and Increased Super Oxide Resistance	Qin Zhou; Yifeng Gao; C.S. Cai et al.	2021	Angewandte Chemie International Edition	140	10.1002/anie.20201[illegible]	[illegible] (2026)

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Effect of Electron-Transport Material on Light-Induced Degradation of Inverted Planar Junction Perovskite Solar Cells	Azat F. Akbulatov; Lyubov A. Frolova; Monroe P. Griffin et al.	2017	Advanced Energy Materials	139	10.1002/aenm.201[illegible]	[illegible] (2026)
A review of stability and progress in tin halide perovskite solar cell	Asim Aftab; Md. Imteyaz Ahmad	2021	Solar Energy	137	10.1016/j.solener.2020.12.065	
Moisture-Resilient Perovskite Solar Cells for Enhanced Stability	Randi Azmi; Shynggys Zhumagali; Helen Bristow et al.	2023	Advanced Materials	134	10.1002/adma.202[illegible]	[illegible] (2026)
Engineering fluorinated-cation containing inverted perovskite solar cells with an efficiency of >21% and improved stability towards humidity	Xiao Wang; Kasparas Rakštys; Kevin S. Jack et al.	2021	Nature Communications	134	10.1038/s41467-020-20272-3	
Review of recent developments and persistent challenges in stability of perovskite solar cells	Billel Salhi; Y.S. Wudil; Mohammad Kamal Hossain et al.	2018	Renewable and Sustainable Energy Reviews	134	10.1016/j.rser.2018.03.058	
Impact of moisture on efficiency-determining electronic processes in perovskite solar cells	Manuel Salado; Lidia Contreras-Bernal; Laura Calió et al.	2017	Journal of Materials Chemistry A	133	10.1039/c7ta02264f	
Stability Improvement of Perovskite Solar Cells by Compositional and Interfacial Engineering	Weiguang Chi; Sanjay K. Banerjee	2021	Chemistry of Materials	132	10.1021/acs.chemm[illegible]31	6.387 (2026)
Encapsulation and Outdoor Testing of Perovskite Solar Cells: Comparing Industrially Relevant Process with a Simplified Lab Procedure	Quiterie Emery; Marko Remec; Gopinath Paramasivam et al.	2022	ACS Applied Materials & Interfaces	132	10.1021/acsami.1c[illegible]	[illegible] (2026)
The Effect of Stoichiometry on the Stability of Inorganic Cesium Lead Mixed-Halide Perovskites Solar Cells	Qingshan Ma; Shujuan Huang; Sheng Chen et al.	2017	The Journal of Physical Chemistry C	131	10.1021/acs.jpcc.7b06268	
Progress and Challenges Toward Effective Flexible Perovskite Solar Cells	Xiongjie Li; Haixuan Yu; Zhirong Liu et al.	2023	Nano-Micro Letters	129	10.1007/s40820-023-01165-8	
A facile molecularly engineered copper (II) phthalocyanine as hole transport material for planar perovskite solar cells with enhanced performance and stability	Guang Yang; Yulong Wang; Jiaju Xu et al.	2016	Nano Energy	126	10.1016/j.nanoen.2016.11.039	
Hermetic seal for perovskite solar cells: An improved plasma enhanced atomic layer deposition encapsulation	Haoran Wang; Yepin Zhao; Zhenyu Wang et al.	2019	Nano Energy	124	10.1016/j.nanoen.2019.104375	
Indigo: A Natural Molecular Passivator for Efficient Perovskite Solar Cells	Junjun Guo; Jianguo Sun; Long Hu et al.	2022	Advanced Energy Materials	124	10.1002/aenm.202[illegible]	[illegible] (2026)
CuSCN as Hole Transport Material with 3D/2D Perovskite Solar Cells	Vinod E. Madhavan; Iwan Zimmermann; Ahmer A.B. Baloch et al.	2019	ACS Applied Energy Materials	124	10.1021/acsaem.9b[illegible]	[illegible] (2026)
Stability of Halide Perovskite Solar Cell Devices: In Situ Observation of Oxygen Diffusion under Biasing	Hee Joon Jung; Daehan Kim; Sung-Kyu Kim et al.	2018	Advanced Materials	123	10.1002/adma.201[illegible]	[illegible] (2026)
Machine learning analysis on stability of perovskite solar cells	Çağla Odabaşı; Ramazan Yıldırım	2019	Solar Energy Materials and Solar Cells	122	10.1016/j.solmat.2019.110284	
Intrinsic and Extrinsic Stability of Formamidinium Lead Bromide Perovskite Solar Cells Yielding High Photovoltage	Neha Arora; M. Ibrahim Dar; Mojtaba Abdi-Jalebi et al.	2016	Nano Letters	122	10.1021/acs.nanolett.6b03455	
Versatility of Carbon Enables All Carbon Based Perovskite Solar Cells to Achieve High Efficiency and High Stability	Xiangyue Meng; Junshuai Zhou; Jie Hou et al.	2018	Advanced Materials	122	10.1002/adma.201[illegible]	[illegible] (2026)
2-CF3-PEAI to eliminate Pb0 traps and form a 2D perovskite layer to enhance the performance and stability of perovskite solar cells	Junjie Zhou; Minghao Li; Siyang Wang et al.	2022	Nano Energy	122	10.1016/j.nanoen.2022.107036	
Encapsulation: The path to commercialization of stable perovskite solar cells	Qianqian Chu; Zhijian Sun; Dong Wang et al.	2023	Matter	120	10.1016/j.matt.2023.08.016	
Ambient Fabrication of Organic–Inorganic Hybrid Perovskite Solar Cells	Yuan Zhang; Ashleigh Kirs; Filip Ambrož et al.	2020	Small Methods	118	10.1002/smtd.202000744	

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Film Grain-Size Related Long-Term Stability of Inverted Perovskite Solar Cells	Chien-Hung Chiang; Chun-Guey Wu	2016	ChemSusChem	117	10.1002/cssc.201600486	5.877 (2026)
Gradated Mixed Hole Transport Layer in a Perovskite Solar Cell: Improving Moisture Stability and Efficiency	Guan-Woo Kim; Gyeongho Kang; Mahdi Malekshahi Byranvand et al.	2017	ACS Applied Materials & Interfaces	116	10.1021/acsami.7b03871	8.387 (2026)
Printable WO3 electron transporting layer for perovskite solar cells: Influence on device performance and stability	Alexandre Gheno; Trang Thi Thu Pham; Catherine Di Bin et al.	2016	Solar Energy Materials and Solar Cells	116	10.1016/j.solmat.2016.10.002	
Barrier reinforcement for enhanced perovskite solar cell stability under reverse bias	Nengxu Li; Zhifang Shi; Chengbin Fei et al.	2024	Nature Energy	114	10.1038/s41560-024-01579-7	
Additive engineering for stable halide perovskite solar cells	Carlos Pereyra; Haibing Xie; Mónica Lira-Cantú	2021	Journal of Energy Chemistry	113	10.1016/j.jechem.2021.01.037	
Mechanochemistry Advances High-Performance Perovskite Solar Cells	Yuzhuo Zhang; Yanju Wang; Xiaoyu Yang et al.	2021	Advanced Materials	112	10.1002/adma.202107420	22.720 (2026)
High Efficiency and Stability of Inverted Perovskite Solar Cells Using Phenethyl Ammonium Iodide-Modified Interface of NiOx and Perovskite Layers	Yuning Liu; Jiaji Duan; Jiankai Zhang et al.	2019	ACS Applied Materials & Interfaces	112	10.1021/acsami.9b18217	8.387 (2026)
Achieving Resistance against Moisture and Oxygen for Perovskite Solar Cells with High Efficiency and Stability	Weiguang Chi; S. Banerjee	2021	Chemistry of Materials	107	10.1021/acs.chemmater.1c00773	6.387 (2026)
Interface degradation of perovskite solar cells and its modification using an annealing-free TiO2 NPs layer	Jian Xiong; Bingchu Yang; Chenghao Cao et al.	2015	Organic Electronics	107	10.1016/j.orgel.2015.12.010	
Boosting mechanical durability under high humidity by bioinspired multisite polymer for high-efficiency flexible perovskite solar cells	Zhihao Li; Zhihao Li; Chunmei Jia et al.	2025	Nature Communications	107	10.1038/s41467-025-57102-3	
Dual Additive for Simultaneous Improvement of Photovoltaic Performance and Stability of Perovskite Solar Cell	In Seok Yang; Nam-Gyu Park	2021	Advanced Functional Materials	105	10.1002/adfm.202100396	14.375 (2026)
Solution processed pristine PDPP3T polymer as hole transport layer for efficient perovskite solar cells with slower degradation	Ashish Dubey; Nirmal Adhikari; Swaminathan Venkatesan et al.	2015	Solar Energy Materials and Solar Cells	105	10.1016/j.solmat.2015.10.008	
Spray reaction prepared FA 1−x Cs x PbI 3 solid solution as a light harvester for perovskite solar cells with improved humidity stability	Xiang Xia; Wenyi Wu; Hongcui Li et al.	2016	RSC Advances	104	10.1039/c5ra23359c	
Amino acid salt-driven planar hybrid perovskite solar cells with enhanced humidity stability	Seong-Cheol Yun; Sunihl Ma; Hyeok-Chan Kwon et al.	2019	Nano Energy	103	10.1016/j.nanoen.2019.02.064	
Degradation and Self-Healing in Perovskite Solar Cells	Blake P. Finkenauer; Akriti Akriti; Ke Ma et al.	2022	ACS Applied Materials & Interfaces	101	10.1021/acsami.2c03825	8.387 (2026)
Recent Progress of All-Bromide Inorganic Perovskite Solar Cells	Guoqing Tong; Luis K. Ono; Yabing Qi	2019	Energy Technology	100	10.1002/ente.201900261	3.826 (2026)
Enhancing stability for organic-inorganic perovskite solar cells by atomic layer deposited Al2O3 encapsulation	Eun Young Choi; Jincheol Kim; Sean Lim et al.	2018	Solar Energy Materials and Solar Cells	99	10.1016/j.solmat.2018.08.016	
Enhanced Ambient Stability of Efficient Perovskite Solar Cells by Employing a Modified Fullerene Cathode Interlayer	Zonglong Zhu; Chu-Chen Chueh; Francis Lin et al.	2016	Advanced Science	97	10.1002/advs.201600027	8.903 (2026)

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Humidity-Assisted Chlorination with Solid Protection Strategy for Efficient Air-Fabricated Inverted CsPbI 3 Perovskite Solar Cells	Sheng Fu; Wenxiao Zhang; Xiaodong Li et al.	2021	ACS Energy Letters	91	10.1021/acsenergy[illegible]	16.7 (2026)
Achieving Long-Term Operational Stability of Perovskite Solar Cells with a Stabilized Efficiency Exceeding 20% after 1000 h	Tae-Youl Yang; Nam Joong Jeon; Hee-Won Shin et al.	2019	Advanced Science	91	10.1002/advs.2019[illegible]	[illegible] (2026)
Graphdiyne-Based Bulk Heterojunction for Efficient and Moisture-Stable Planar Perovskite Solar Cells	Hongshi Li; Rui Zhang; Yusheng Li et al.	2018	Advanced Energy Materials	90	10.1002/aenm.201[illegible]	[illegible] (2026)
Long-term stability of organic–inorganic hybrid perovskite solar cells with high efficiency under high humidity conditions	Yue Sun; Yihui Wu; Xiang Fang et al.	2016	Journal of Materials Chemistry A	89	10.1039/c6ta08117g	
Tetrabutylammonium cations for moisture-resistant and semitransparent perovskite solar cells	Isabella Poli; Salvador Eslava; Petra J. Cameron	2017	Journal of Materials Chemistry A	87	10.1039/c7ta06735f	
Textured CH3NH3PbI3 thin film with enhanced stability for high performance perovskite solar cells	Mingzhu Long; Tiankai Zhang; Houyu Zhu et al.	2017	Nano Energy	86	10.1016/j.nanoen.2017.02.002	
Stability Enhancement in Perovskite Solar Cells with Perovskite/Silver–Graphene Composites in the Active Layer	Tahmineh Mahmoudi; Yousheng Wang; Yoon-Bong Hahn	2018	ACS Energy Letters	86	10.1021/acsenergy[illegible]	16.7 (2026)
Reviewing and understanding the stability mechanism of halide perovskite solar cells	Caixin Zhang; Tao Shen; Dan Guo et al.	2020	InfoMat	84	10.1002/inf2.12104	
Double Barriers for Moisture Degradation: Assembly of Hydrolysable Hydrophobic Molecules for Stable Perovskite Solar Cells with High Open-Circuit Voltage	Pengfei Guo; Qian Ye; Chen Liu et al.	2020	Advanced Functional Materials	84	10.1002/adfm.2020[illegible]	[illegible] (2026)
Spontaneous grain polymerization for efficient and stable perovskite solar cells	Xiaodong Li; Wenxiao Zhang; Wenjun Zhang et al.	2019	Nano Energy	84	10.1016/j.nanoen.2019.02.009	
Glass-to-glass encapsulation with ultraviolet light curable epoxy edge sealing for stable perovskite solar cells	Easwaramoorthi Ramasamy; Vaithinathan Karthikeyan; Kadusu Rameshkumar et al.	2019	Materials Letters	83	10.1016/j.matlet.2019.04.082	
Composite Encapsulation Enabled Superior Comprehensive Stability of Perovskite Solar Cells	Yifan Lv; Hui Zhang; Ruqing Liu et al.	2020	ACS Applied Materials & Interfaces	82	10.1021/acsami.0c[illegible]	[illegible] (2026)
Degradation mechanism of planar-perovskite solar cells: correlating evolution of iodine distribution and photocurrent hysteresis	Riski Titian Ginting; Mi-Kyoung Jeon; Kwang-Jae Lee et al.	2017	Journal of Materials Chemistry A	80	10.1039/c6ta09202k	
Stability Issues of Perovskite Solar Cells: A Critical Review	Shahriyar Safat Dipta; Ashraf Uddin	2021	Energy Technology	79	10.1002/ente.2021[illegible]	[illegible] (2026)
Advancement in CsPbBr3 inorganic perovskite solar cells: Fabrication, efficiency and stability	Naveen Kumar; Jyoti Rani; Rajnish Kurchania	2021	Solar Energy	79	10.1016/j.solener.2021.04.042	
3D/2D multidimensional perovskites: Balance of high performance and stability for perovskite solar cells	Fei Zhang; Dong Hoe Kim; Kai Zhu	2018	Current Opinion in Electrochemistry	79	10.1016/j.coelec.20[illegible]	[illegible] (2026)
Electrospray-Assisted Fabrication of Moisture-Resistant and Highly Stable Perovskite Solar Cells at Ambient Conditions	Shalinee Kavadiya; Dariusz M. Niedzwiedzki; Su Huang et al.	2017	Advanced Energy Materials	76	10.1002/aenm.201[illegible]	[illegible] (2026)

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Printing High-Efficiency Perovskite Solar Cells in High-Humidity Ambient Environment—An In Situ Guided Investigation	W.K. Fong; Hanlin Hu; Zhiwei Ren et al.	2021	Advanced Science	75	10.1002/advs.2020…	8.339 (2026)
Self-healing perovskite solar cells	Yue Yu; Fu Zhang; Hua Yu	2020	Solar Energy	75	10.1016/j.solener.2020.09.018	
Effect of relative humidity during the preparation of perovskite solar cells: Performance and stability	Isabel Mesquita; Luísa Andrade; Adélio Mendes	2020	Solar Energy	72	10.1016/j.solener.2020.02.052	
Enhanced Moisture Stability by Butyldimethylsulfonium Cation in Perovskite Solar Cells	Bo-Hyung Kim; Maengsuk Kim; Jun Hee Lee et al.	2019	Advanced Science	71	10.1002/advs.2019…	8.9… (2026)
Excellent Moisture Stability and Efficiency of Inverted All-Inorganic CsPbIBr2 Perovskite Solar Cells through Molecule Interface Engineering	Shuzhang Yang; Liang Wang; Liguo Gao et al.	2020	ACS Applied Materials & Interfaces	70	10.1021/acsami.9b…	8.3… (2026)
Amphiphilic Fullerenes Employed to Improve the Quality of Perovskite Films and the Stability of Perovskite Solar Cells	Qingxia Fu; Shuqin Xiao; Xianglan Tang et al.	2019	ACS Applied Materials & Interfaces	70	10.1021/acsami.9b…	8.… (2026)
From Exceptional Properties to Stability Challenges of Perovskite Solar Cells	Somayeh Gholipour; Michael Saliba	2018	Small	69	10.1002/smll.201802385	
Aromatic Alkylammonium Spacer Cations for Efficient Two-Dimensional Perovskite Solar Cells with Enhanced Moisture and Thermal Stability	Deepak Thrithamarassery Gangadharan; Yujie Han; Ashish Dubey et al.	2018	Solar RRL	68	10.1002/solr.201700215	
Thermal and Humidity Stability of Mixed Spacer Cations 2D Perovskite Solar Cells	Huayang Yu; Yulin Xie; Jia Zhang et al.	2021	Advanced Science	67	10.1002/advs.2020…	8.… (2026)
Chlorine-terminated MXene quantum dots for improving crystallinity and moisture stability in high-performance perovskite solar cells	Xiao Liu; Zhiang Zhang; Jinkun Jiang et al.	2021	Chemical Engineering Journal	64	10.1016/j.cej.20211…1…	(2026)
Stability and Performance Enhancement of Perovskite Solar Cells: A Review	Maria Khalid; Tapas K. Mallick	2023	Energies	62	10.3390/en161040…	…847 (2026)
Progress on the stability and encapsulation techniques of perovskite solar cells	Ling Xiang; Fangliang Gao; Yunxuan Cao et al.	2022	Organic Electronics	61	10.1016/j.orgel.2022.106515	
Encapsulating perovskite solar cells for long-term stability and prevention of lead toxicity	Shahriyar Safat Dipta; Md. Arifur Rahim; Ashraf Uddin	2024	Applied Physics Reviews	61	10.1063/5.019715…	…1.309 (2026)
Recent Progress on the Stability of Perovskite Solar Cells in a Humid Environment	Mengying Li; Haibo Li; Jing Fu et al.	2020	The Journal of Physical Chemistry C	60	10.1021/acs.jpcc.0c08019	
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Homeopathic Perovskite Solar Cells: Effect of Humidity during Fabrication on the Performance and Stability of the Device	Lidia Contreras-Bernal; Clara Aranda; Marta Vallés-Pelarda et al.	2018	The Journal of Physical Chemistry C	56	10.1021/acs.jpcc.8b01558	
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Enhanced moisture stability of MAPbI3 perovskite solar cells through Barium doping	Jitendra Bahadur; Amir H. Ghahremani; Sunil Gupta et al.	2019	Solar Energy	51	10.1016/j.solener.2019.08.033	
Reduced Graphene Oxide Improves Moisture and Thermal Stability of Perovskite Solar Cells	Hui-Seon Kim; Bowen Yang; Minas M. Stylianakis et al.	2020	Cell Reports Physical Science	50	10.1016/j.xcrp.202[illegible]	[illegible] (2026)
A simple method to evaluate the effectiveness of encapsulation materials for perovskite solar cells	Timothy J. Wilderspin; Francesca De Rossi; Trystan Watson	2016	Solar Energy	48	10.1016/j.solener.2016.09.038	
Simultaneously enhanced moisture tolerance and defect passivation of perovskite solar cells with cross-linked grain encapsulation	Ke Xiao; Qiaolei Han; Yuan Gao et al.	2020	Journal of Energy Chemistry	46	10.1016/j.jechem.2020.08.020	
Protecting Perovskite Solar Cells against Moisture-Induced Degradation with Sputtered Inorganic Barrier Layers	Ramez Hosseinian Ahangharnejhad; Zhaoning Song; Tamanna Mariam et al.	2021	ACS Applied Energy Materials	46	10.1021/acsaem.1c[illegible]	[illegible] (2026)
Role of interface in stability of perovskite solar cells	Chris Manspeaker; Swaminathan Venkatesan; Alex Zakhidov et al.	2016	Current Opinion in Chemical Engineering	45	10.1016/j.coche.20[illegible]	[illegible] (2026)
Hysteresis and Instability Predicted in Moisture Degradation of Perovskite Solar Cells	Kelvin J. Xu; Ryan Taoran Wang; Fan Xu et al.	2020	ACS Applied Materials & Interfaces	38	10.1021/acsami.0c[illegible]	[illegible] (2026)
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Moisture stability in nanostructured perovskite solar cells	Mahdi Malekshahi Byranvand; Ali Nemati Kharat; Nima Taghavinia	2018	Materials Letters	31	10.1016/j.matlet.2018.10.029	
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Degradation Kinetics of Inverted Perovskite Solar Cells	Mejd Alsari; Andrew J. Pearson; Jacob Tse-Wei Wang et al.	2018	arXiv preprint	0	(link)	
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Effect of Ion Migration Induced Electrode Degradation on the Operational Stability of Perovskite Solar Cells	Boris Rivkin; Paul Fassl; Qing Sun et al.	2019	arXiv preprint	0	(link)	
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Quantifying Perovskite Solar Cell Degradation via Machine Learning from Spatially Resolved Multimodal Luminescence Time Series	Giulio Barletta; Simon Ternes; Saif Ali et al.	2026	arXiv preprint	0	(link)	
Thermal Degradation Mechanisms and Stability Enhancement Strategies in Perovskite Solar Cells: A Review	Arghya Paul; Kanak Raj; Prince Raj Lawrence Raj et al.	2025	arXiv preprint	0	(link)	
Architecture Optimization Dramatically Improves Reverse Bias Stability in Perovskite Solar Cells: A Role of Polymer Hole Transport Layers	Fangyuan Jiang; Yangwei Shi; Tanka R. Rana et al.	2023	arXiv preprint	0	(link)	
Improving efficiency and stability for perovskite solar cell with diethylene glycol dimethacrylate modification	Xiaodan Wei; Shuyuan Zhang; Boyang Ma et al.	2025	arXiv preprint	0	(link)	

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Boosting Perovskite Solar Cell Stability: Dual Protection with Ultrathin Plasma Polymer Passivation Layers	Mahmoud Nabil; Lidia Contreras-Bernal; Gloria P. Moreno-Martinez et al.	2024	arXiv preprint	0	(link)	
Disentangling the origin of degradation in perovskite solar cells via optical imaging and Bayesian inference	Akash Dasgupta; Robert D. J. Oliver; Manuel Kober-Czerny et al.	2026	arXiv preprint	0	(link)	
Analysis of degradation in perovskite solar cells through physics-based machine learning	Kjeld O. Jensen; Gemma Giliberti; Aldo Di Carlo et al.	2026	arXiv preprint	0	(link)	
Critical role of water in defect aggregation and chemical degradation of perovskite solar cells	Yun-Hyok Kye; Chol-Jun Yu; Un-Gi Jong et al.	2017	arXiv preprint	0	(link)	
Ionic Charge Imbalance in Perovskite Solar Cells	Dhyana Sivadas; Abhimanyu Singareddy; Chittiboina vinod et al.	2023	arXiv preprint	0	(link)	
First-principles studies of oxygen interstitial dopants in RbPbI$_3$ halide for perovskite solar cells	Chongyao Yang; Wei Wu; Kwang-Leong Choy	2023	Modelling and Simulation in Materials Science and Engineering, 32, 015007 (2023)	0	10.1088/1361-651X/ad104f	
Efficient indoor p-i-n hybrid perovskite solar cells using low temperature solution processed NiO as hole extraction layers	Lethy Krishnan Jagadamma; Oskar Blaszczyk; Muhammad T. Sajjad et al.	2019	Solar Energy Materials and Solar Cells, 201, 110071, 2019	0	10.1016/j.solmat.2019.110071	
Enhancing Intrinsic Stability of Hybrid Perovskite Solar Cell by Strong, yet Balanced, Electronic Coupling	Fedwa El-Mellouhi; El Tayeb Bentria; Sergey N Rashkeev et al.	2016	arXiv preprint	0	(link)	
Charge transport layer dependent electronic band bending in perovskite solar cells and its correlation to device degradation	Junseop Byeon; Jutae Kim; Ji-Young Kim et al.	2019	arXiv preprint	0	(link)	
Nitrobenzene as Additive to Improve Reproducibility and Degradation Resistance of Highly Efficient Methylammonium-Free Inverted Perovskite Solar Cells	Apostolos Ioakeimidis; Stelios. A. Choulis	2020	Materials 2020, 13, 3289	0	10.3390/ma13153289	
Ultrathin plasma polymer passivation of perovskite solar cells for improved stability and reproducibility	Jose M. Obrero-Perez; Lidia Contreras-Bernal+; Fernando Nunez-Galvez et al.	2022	arXiv preprint	0	(link)	
Material selection method for a perovskite solar cell design based on the genetic algorithm	Eungkyun Kim; Indranil Bhattacharya	2020	arXiv preprint	0	(link)	
Low Frequency Carrier Kinetics in Perovskite Solar Cells	Vinod K. Sangwan; Menghua Zhu; Sarah Clark et al.	2019	arXiv preprint	0	10.1021/acsami.9b03884	
Multimodal operando microscopy reveals that interfacial chemistry and nanoscale performance disorder dictate perovskite solar cell stability	Kyle Frohna; Cullen Chosy; Amran Al-Ashouri et al.	2024	arXiv preprint	0	(link)	
Bayesian parameter estimation for characterising mobile ion vacancies in perovskite solar cells	Samuel G. McCallum; Oliver Nicholls; Kjeld O. Jensen et al.	2023	arXiv preprint	0	(link)	
Enhanced Power Point Tracking for High Hysteresis Perovskite Solar Cells: A Galvanostatic Approach	Emilio J. Juarez-Perez; Cristina Momblona; Roberto Casas et al.	2023	Cell Reports Physical Science, 2024, 5, 101885	0	10.1016/j.xcrp.2024.101885	
Quantification of mobile ions in perovskite solar cells with thermally activated ion current measurements	Moritz C. Schmidt; Agustin O. Alvarez; Riccardo Pallotta et al.	2025	arXiv preprint	0	(link)	
Photo Stabilization of p-i-n Perovskite Solar Cells with Bathocuproine: MXene	Anastasia Yusheva; Danila Saranin; Dmitry Muratov et al.	2023	Small, 2022	0	(link)	
Water-resistant hybrid perovskite solar cell – drop triboelectric energy harvester	Fernando Nunez-Galvez; Xabier Garcia-Casas; Lidia Contreras Bernal et al.	2024	Nano Energy, vol. 148, 2026, 111678	0	10.1016/j.nanoen.2025.111678	
Deposition of Reduced Graphene Oxide Thin Film by Spray Pyrolysis Method for Perovskite Solar Cell	Manoj Pandey; Dipendra Hamal; Deepak Subedi et al.	2022	arXiv preprint	0	10.3126/jnphyssoc.v7i3.42193	

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Simultaneous Optimization of Efficiency and Degradation in Tunable HTL-Free Perovskite Solar Cells with MWCNT-Integrated Back Contact Using a Machine Learning-Derived Polynomial Regressor	Ihtesham Ibn Malek; Hafiz Imtiaz; Samia Subrina	2025	arXiv preprint	0	(link)	
Influence of Phase Segregation on the Hysteresis of Perovskite Solar Cells	Abhimanyu Singareddy; Pradeep R. Nair	2024	arXiv preprint	0	(link)	
Pulsatile therapy for perovskite solar cells	Kiwan Jeong; Junseop Byeon; Jihun Jang et al.	2020	arXiv preprint	0	(link)	
LLM tools in the prediction of the stability of perovskite solar cells	S. Frenkel; V. Zakharov; E. A. Katz	2025	arXiv preprint	0	(link)	
Efficiency limits of Perovskite Solar Cells with Transition Metal Oxides as Hole Transport Layers	Dhyana Sivadas; Swasti Bhatia; Pradeep Nair	2021	arXiv preprint	0	10.1063/5.0059221	
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Employing surfactant-assisted hydrothermal synthesis to control CuGaO2 nanoparticle formation and improved carrier selectivity of perovskite solar cells	Ioannis T. Papadas; Achilleas Savva; Apostolos Ioakeimidis et al.	2018	Materials Today Energy, Volume 8, Pages 57-64, 2018	0	10.1016/j.mtener.2018.03.003	
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A Design Based on Staircase Band Alignment of Electron Transport Layer for Improving Performance and Stability in Planar Perovskite Solar Cells	Shang-Hsuan Wu; Ming-Yi Lin; Sheng-Hao Chang et al.	2017	arXiv preprint	0	(link)	
Improved evaluation of deep-level transient spectroscopy on perovskite solar cells reveals ionic defect distribution	Sebastian Reichert; Jens Flemming; Qingzhi An et al.	2019	Phys. Rev. Applied 13, 034018 (2020)	0	10.1103/PhysRevApplied.13.034018	
Automated and Scalable SEM Image Analysis of Perovskite Solar Cell Materials via a Deep Segmentation Framework	Jian Guo Pan; Lin Wang; Xia Cai	2025	arXiv preprint	0	(link)	
Double-side Integration of the Fluorinated Self-Assembling Monolayers for Enhanced Stability of Inverted Perovskite Solar Cells	Ekaterina A. Ilicheva; Polina K. Sukhorukova; Lev O. Luchnikov et al.	2024	arXiv preprint	0	(link)	
Unraveling UV Stability in Metal Halide Perovskites: From Degradation Mechanisms to Molecular Passivation	Xin Wen; Zhiyi Yao; Wenzhuo Li et al.	2025	arXiv preprint	0	(link)	
Impact of Doping Spiro-OMeTAD with Li-TFSI, FK209, and tBP on the Performance of Perovskite Solar Cells	Mehran Hosseinzadeh Dizaj	2024	arXiv preprint	0	10.22036/org.chem.2024.494577.1370	

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Dopant-free molecular hole transport material that mediates a 20% power conversion efficiency in a perovskite solar cell	Yang Cao; Yunlong Li; Thomas Morrissey et al.	2019	arXiv preprint	0	(link)	
Comparison of Electroluminescence and Photoluminescence Imaging of Mixed-Cation Mixed-Halide Perovskite Solar Cells at Low Temperatures	Hurriyet Yuce-Cakir; Haoran Chen; Isaac Ogunniranye et al.	2025	arXiv preprint	0	(link)	
Operando multidimensional spectroscopy reveals A-site-dependent carrier cooling in perovskite solar cells	Edoardo Amarotti; Luca Bolzonello; Qi Shi et al.	2026	arXiv preprint	0	(link)	
La-Doped BaSnO$_3$ Electron Transport Layer for Perovskite Solar Cells	Chang Woo Myung; Geunsik Lee; Kwang S. Kim	2018	arXiv preprint	0	(link)	
NiOx passivation in perovskite solar cells: from surface reactivity to device performance	John Mohanraj; Bipasa Samanta; Osbel Almora et al.	2024	arXiv preprint	0	(link)	
Phenothiazine-Based Self-Assembled Monolayer with Thiophene Head Groups Minimizes Buried Interface Losses in Tin Perovskite Solar Cells	Valerio Stacchini; Madineh Rastgoo; Mantas Marčinskas et al.	2025	arXiv preprint	0	10.1002/aenm.202500841	
Exploring the Way to Approach the Efficiency Limit of Perovskite Solar Cells by Drift-Diffusion Model	Xingang Ren; Zishuai Wang; Wei E. I. Sha et al.	2017	ACS Photonics, 2017, 4(4), 934-942	0	10.1021/acsphotonics.6b01043	
Triphenylamine-based interlayer with carboxyl anchoring group for tuning of charge collection interface in stabilized p-i-n perovskite solar cells and modules	P. K. Sukhorukova; E. A. Ilicheva; P. A. Gostishchev et al.	2023	arXiv preprint	0	(link)	
Ion Induced Passivation of Grain Boundaries in Perovskite Solar Cells	Vikas Nandal; Pradeep R. Nair	2018	arXiv preprint	0	10.1063/1.5082967	
Super-droplet-repellent carbon-based printable perovskite solar cells	Cuc Thi Kim Mai; Janne Halme; Heikki A. Nurmi et al.	2024	arXiv preprint	0	(link)	
An eco-friendly passivation strategy of resveratrol for highly efficient and antioxidative perovskite solar cells	Xianhu Wu; Jieyu Bi; Guanglei Cui et al.	2024	arXiv preprint	0	(link)	
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Improved Performance and Reliability of p-i-n Perovskite Solar Cells via Doped Metal Oxides	Achilleas Savva; Ignasi Burgues-Ceballos; Stelios A. Choulis	2016	Advanced Energy Materials 2016, 1600285	0	10.1002/aenm.201600285	
Influences of Dielectric Constant and Scan Rate to Hysteresis Effect in Perovskite Solar Cell: Simulation and Experimental Analyses	Jun-Yu Huang; You-Wei Yang; Wei-Hsuan Hsu et al.	2021	Scientific Reports, 12, 7927, 2022	0	10.1038/s41598-022-11899-x	
Optimization and Performance Evaluation of Cs$_2$CuBiCl$_6$ Double Perovskite Solar Cell for Lead-Free Photovoltaic Applications	Syeda Anber Urooj Wasti; Sundus Naz; Ammara Sattar et al.	2025	arXiv preprint	0	(link)	
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Surface Treatment of Cu:NiOx Hole-Transporting Layer Using \b{eta}-Alanine for Hysteresis-Free and Thermally Stable Inverted Perovskite Solar Cells	Fedros Galatopoulos; Ioannis T. Papadas; Apostolos Ioakeimidis et al.	2020	Nanomaterials 2020, 10(10), 1961	0	10.3390/nano10101961	
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A green solvent system for precursor phase-engineered sequential deposition of stable formamidinium lead triiodide for perovskite solar cells	Benjamin M. Gallant; Philippe Holzhey; Joel A. Smith et al.	2024	arXiv preprint	0	(link)	
Impact of interface defects on the band alignment and performance of TiO$_2$/MAPI/Cu$_2$O perovskite solar cells	Nicolae Filipoiu; Marina Cuzminschi; Mihaela Cosinschi et al.	2024	Physica Scripta 101, 265906 (2026)	0	10.1088/1402-4896/ae7cef	
Efficient Passivation of Surface Defects by Lewis Base in Lead-free Tin-based Perovskite Solar Cells	Hejin Yan; Bowen Wang; Xuefei Yan et al.	2022	Materials Today Energy (2022)	0	10.1016/j.mtener.2022.101038	
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Deep-level transient spectroscopy of the charged defects in p-i-n perovskite solar cells induced by light-soaking	A. A. Vasilev; D. S. Saranin; P. A. Gostishchev et al.	2022	arXiv preprint	0	(link)	
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Combined DFT, SCAPS-1D, and wxAMPS frameworks for design optimization of efficient Cs2BiAgI6-based perovskite solar cells with different charge transport layers	M. Khalid Hossain; A. A. Arnab; Ranjit C. Das et al.	2022	RSC Advance 12 (2022) 35002-35025	0	10.1039/D2RA06734J	
A Comprehensive Computational Photovoltaic Study of Lead-free Inorganic NaSnCl$_3$-based Perovskite Solar Cell: Effect of Charge Transport Layers and Material Parameters	Md Tashfiq Bin Kashem; Sadia Anjum Esha	2025	arXiv preprint	0	(link)	
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Structural Stability and Optoelectronic Properties of Tetragonal MAPbI3 under Strain	Lei Guo; Gao Xu; Gang Tang et al.	2020	Nanotechnology 31, 225204 (2020)	0	10.1088/1361-6528/ab7679	
Stability Engineering of Halide Perovskite via Machine Learning	Zhenzhu Li; Qichen Xu; Qingde Sun et al.	2018	arXiv preprint	0	(link)	
Nanoparticulate Metal Oxide Top Electrode Interface Modification Improves the Thermal Stability of Inverted Perovskite Photovoltaics	Ioannis T. Papadas; Fedros Galatopoulos; Gerasimos S. Armatas et al.	2019	Nanomaterials 2019, 9(11), 1616	0	10.3390/nano9111616	
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Enhanced stability of 2D organic-inorganic halide perovskites by doping and heterostructure engineering	Rahul Singh; Prashant Singh; Ganesh Balasubramanian	2018	arXiv preprint	0	(link)	
Influence of halide composition on the structural, electronic, and optical properties of mixed CH$_3$NH$_3$Pb(I$_{1-x}$Br$_x$)$_3$ perovskites calculated using the virtual crystal approximation method	Un-Gi Jong; Chol-Jun Yu; Nam-Hyok Kim et al.	2016	Phys. Rev. B 94, 125139 (2016)	0	10.1103/PhysRevB.94.125139	
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Protective Coating Interfaces for perovskite Solar Cell Materials: A first Principles Study	Azimatu Fangnon; Marc Dvorak; Ville Havu et al.	2022	arXiv preprint	0	10.1021/acsami.1c21785	
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Multi-site mixing and entropy stabilization of CsPbI$_{3}$ with potential application in photovoltaics	Namitha Anna Koshi; Krishnamohan Thekkepat; Doh-Kwon Lee et al.	2025	arXiv preprint	0	(link)	
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Transmission electron microscopy of organic-inorganic hybrid perovskites: myths and truths	Shulin Chen; Ying Zhang; Jinjin Zhao et al.	2020	Science Bulletin. 65, 1643-1649 (2020)	0	10.1016/j.scib.2020.05.020	
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Multiscale Models For Perovskite Optimisation	Philippe Baranek; James P. Connolly; Antoine Gissler et al.	2025	EUPVSEC 2025 42nd European Photovoltaic Solar Energy Conference and Exhibition, WIP Renewable Energies, Sep 2025, Bilbao, Spain	0	(link)	

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Lattice matched heterogeneous nucleation eliminate defective buried interface in halide perovskites	Paramvir Ahlawat; Cecilia Clementi; Felix Musil et al.	2024	arXiv preprint	0	(link)	
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Resolving the Hydrophobicity of Me-4PACz Hole Transport Layer for High-Efficiency Inverted Perovskite Solar Cells	Kashimul Hossain; Ashish Kulkarni; Urvashi Bothra et al.	2023	arXiv preprint	0	(link)	
First-principles study on the chemical decomposition of inorganic perovskites \ce{CsPbI3} and \ce{RbPbI3} at finite temperature and pressure	Un-Gi Jong; Chol-Jun Yu; Yun-Hyok Kye et al.	2018	arXiv preprint	0	(link)	
Compound Defects in Halide Perovskites: A First-Principles Study of CsPbI$_3$	Haibo Xue; José Manuel Vicent-Luna; Shuxia Tao et al.	2022	arXiv preprint	0	(link)	
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Recent advances in La2NiMnO6 Double Perovskites for various applications; Challenges and opportunities	Suresh Chandra Baral; P. Maneesha; E. G. Rini et al.	2023	arXiv preprint	0	(link)	
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Fully printed flexible perovskite solar modules with improved energy alignment by tin oxide surface modification	Lirong Dong; Shudi Qiu; Jose Garcia Cerrillo et al.	2024	arXiv preprint	0	(link)	
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Precise Control of Process Parameters for >23% Efficiency Perovskite Solar Cells in Ambient Air Using an Automated Device Acceleration Platform	Jiyun Zhang; Anastasia Barabash; Tian Du et al.	2024	arXiv preprint	0	(link)	
Dimethylammonium additives alter both vertical and lateral composition in halide perovskite semiconductors	Sarthak Jariwala; Rishi Kumar; Giles E. Eperon et al.	2021	arXiv preprint	0	(link)	
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Substitution of Lead with Tin Suppresses Ionic Transport in Halide Perovskite Optoelectronics	Krishanu Dey; Dibyajyoti Ghosh; Matthew Pilot et al.	2023	arXiv preprint	0	(link)	
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Efficient dataset generation for machine learning perovskite alloys	Henrietta Homm; Jarno Laakso; Patrick Rinke	2025	Physical Review Materials, 9(5), 053802 (2025)	0	10.1103/PhysRevMaterials.9.053802	
Dynamic Nanodomains Dictate Macroscopic Properties in Lead Halide Perovskites	Milos Dubajic; James R. Neilson; Johan Klarbring et al.	2024	arXiv preprint	0	10.1038/s41565-025-01917-0	

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A closed-loop AI framework for hypothesis-driven and interpretable materials design	Kangyu Ji; Tianran Liu; Fang Sheng et al.	2025	arXiv preprint	0	(link)	
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Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics	Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.	2018	The Journal of Physical Chemistry Letters	269	10.1021/acs.jpclett.8b00902	
Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide	Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur et al.	2020	npj Computational Materials	185	10.1038/s41524-020-00367-7	
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Growth Mechanism and Origin of High sp3 Content in Tetrahedral Amorphous Carbon	Miguel A. Caro; Volker L. Deringer; Jari Koskinen et al.	2018	Physical Review Letters	178	10.1103/physrevlett.120.166101	
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Study of Li atom diffusion in amorphous Li3PO4 with neural network potential	Wenwen Li; Yasunobu Ando; Emi Minamitani et al.	2017	The Journal of Chemical Physics	154	10.1063/1.4997242	
Impact of the Local Environment on Li Ion Transport in Inorganic Components of Solid Electrolyte Interphases	Taiping Hu; Jianxin Tian; Fuzhi Dai et al.	2022	Journal of the American Chemical Society	107	10.1021/jacs.2c11521	
Thermal conductivity modeling using machine learning potentials: application to crystalline and amorphous silicon	Xin Qian; Shiyu Peng; Xiaobo Li et al.	2019	Materials Today Physics	103	10.1016/j.mtphys.2019.100140	
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Ultralow and glass-like lattice thermal conductivity in crystalline BaAg2Te2: Strong fourth-order anharmonicity and crucial diffusive thermal transport	Zezhu Zeng; Cunzhi Zhang; Hulei Yu et al.	2021	Materials Today Physics	37	10.1016/j.mtphys.2021.100487	
Dependence of a cooling rate on structural and vibrational properties of amorphous silicon: A neural network potential-based molecular dynamics study	Wenwen Li; Yasunobu Ando	2019	The Journal of Chemical Physics	35	10.1063/1.5114652	
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Applications of machine-learning interatomic potentials for modeling ceramics, glass, and electrolytes: A review	Shingo Urata; Marco Bertani; Alfonso Pedone	2024	Journal of the American Ceramic Society	32	10.1111/jace.19934	
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Ab initio molecular dynamics and high-dimensional neural network potential study of VZrNbHfTa melt	I. A. Balyakin; A. A. Yuryev; . . et al.	2020	Journal of Physics Condensed Matter	20	10.1088/1361-648x/ab6f87	
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Binding energies and sticking coefficients of H2 on crystalline and amorphous CO ice	Germán Molpeceres; Viktor Zaverkin; Naoki Watanabe et al.	2021	Astronomy and Astrophysics	18	10.1051/0004-6361/202040023	5.028 (2026)
Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics	Yanzhou Wang; Zheyong Fan; Ping Qian et al.	2025	Physical review. B./Physical review. B	17	10.1103/physrevb.111.094205	
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Atomic Structure of Na 4 P 2 S 7 Glass Solid Electrolyte: Fine-Tuning Machine Learning Potentials for Enhanced Accuracy	Marco Bertani; Alfonso Pedone	2025	The Journal of Physical Chemistry C	12	10.1021/acs.jpcc.5c01857	

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Machine Learning Interatomic Potential for Modeling the Mechanical and Thermal Properties of Naphthyl-Based Nanotubes	Hugo X. Rodrigues; Hudson R. Armando; Daniel A. da Silva et al.	2025	Journal of Chemical Theory and Computation	12	10.1021/acs.jctc.4c01578	
Neural Network Force Fields for Metal Growth Based on Energy Decompositions	Qin Hu; Mouyi Weng; Xin Chen et al.	2020	The Journal of Physical Chemistry Letters	12	10.1021/acs.jpclett.9b03780	
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Accurate prediction of structural and mechanical properties on amorphous materials enabled through machine-learning potentials: A case study of silicon nitride	Ganesh Kumar Nayak; Prashanth Srinivasan; Juraj Todt et al.	2025	Computational Materials Science	11	10.1016/j.commatsci.[illegible]	[illegible] (2026)
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Atomistic Study of the Configurational Entropy and the Fragility of Supercooled Liquid GeTe	Yuhan Chen; Davide Campi; Marco Bernasconi et al.	2024	Advanced Functional Materials	9	10.1002/adfm.2023[illegible]	[illegible] (2026)
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Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics.	Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.	2018	arXiv (Cornell University)	6	10.17863/cam.28044	
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Study of vacancy ordering and the boson peak in metastable cubic Ge-Sb-Te using machine learning potentials	Young-Jae Choi; Minjae Ghim; Seung-Hoon Jhi	2024	Physical Review Materials	6	10.1103/physrevmaterials.8.013606	
Atomistic Simulations of the Crystallization of Amorphous GeTe Nanoparticles	Debdipto Acharya; Omar Abou El Kheir; Simone Perego et al.	2024	The Journal of Physical Chemistry C	6	10.1021/acs.jpcc.4c05157	
Multikernel similarity-based clustering of amorphous systems and machine-learned interatomic potentials by active learning	Firas Shuaib; Guido Ori; Philippe Thomas et al.	2024	Journal of the American Ceramic Society	5	10.1111/jace.20128	
Crystallization kinetics in Ge-rich Ge x Te alloys from large scale simulations with a machine-learned interatomic potential	Dario Baratella; Omar Abou El Kheir; Marco Bernasconi	2024	Acta Materialia	5	10.1016/j.actamat.20[illegible]	[illegible] (2026)
Amorphous Zirconia-doped Tantala modeling and simulations using explicit multi-element spectral neighbor analysis machine learning potentials (EME-SNAP)	Jun Jiang; Xiangguo Li; A. Mishkin et al.	2023	Physical Review Materials	5	10.1103/physrevmaterials.7.045602	
Temperature-density dependent hidden order of amorphous carbon	Yutao Liu; Tinghong Gao; Qingquan Xiao et al.	2025	Communications Physics	5	10.1038/s42005-025-02320-w	5.555 (2026)
The ab initio amorphous materials database: Empowering machine learning to decode diffusivity	Hui Zheng; Eric Sivonxay; Max C. Gallant et al.	2024	arXiv (Cornell University)	5	10.48550/arxiv.2402.00177	
Mechanisms of nanodiamond and amorphous diamond-like carbon formation following room temperature compression of C60	Hendrik Heimes; Ethan P. Turner; Alan Salek et al.	2025	Diamond and Related Materials	5	10.1016/j.diamond.[illegible]	[illegible] (2026)
Non-Monotonic Ion Conductivity in Lithium-Aluminum-Chloride Glass Solid-State Electrolytes Explained by Cascading Hopping	Beomgyu Kang; Jinqiu Yu; Shinji Saito et al.	2025	Advanced Science	5	10.1002/advs.20250[illegible]	[illegible] (2026)
A high-efficiency neuroevolution potential for tobermorite and calcium silicate hydrate systems with ab initio accuracy	Xiao Xu; Shijie Wang; Haifeng Qin et al.	2025	Cement and Concrete Research	5	10.1016/j.cemconres.[illegible]	[illegible] (2026)
Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures	Zhuo Chen; Yuejin Yuan; Yanzhou Wang et al.	2025	Advanced Electronic Materials	5	10.1002/aelm.20250[illegible]	[illegible] (2026)
Machine learning potential for predicting thermal conductivity of -phase and amorphous tantalum nitride	Zhicheng Zong; Yangjun Qin; Jiahong Zhan et al.	2025	International Journal of Heat and Mass Transfer	4	10.1016/j.ijheatmasstransfer.2025.128155	
Molecular Dynamics Study of Silicon Carbide Using an Ab Initio-Based Neural Network Potential: Effect of Composition and Temperature on Crystallization Behavior	Jinyoung Lim; Youngseon Shim; Jaehong Park et al.	2023	The Journal of Physical Chemistry C	4	10.1021/acs.jpcc.3c04224	
Slowly quenched, high pressure glassy B2O3 at DFT accuracy	Debendra Meher; Nikhil V. S. Avula; Sundaram Balasubramanian	2025	The Journal of Chemical Physics	4	10.1063/5.0240030	
Nature of the amorphous-amorphous interfaces in solid-state batteries revealed using machine-learned interatomic potentials	Chuhong Wang; Muratahan Aykol; Tim Mueller	2023	ChemRxiv	3	10.26434/chemrxiv-2023-frr79-v2	0.467 (2026)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Viscosity, breakdown of Stokes–Einstein relation, and dynamical heterogeneity in supercooled liquid Ge2Sb2Te5 from simulations with a neural network potential	Simone Marcorini; Rocco Pomodoro; Omar Abou El Kheir et al.	2025	The Journal of Chemical Physics	3	10.1063/5.0282855	
Anisotropic response under shock compression in refractory high-entropy alloy via machine-learning potential	Hongyang Liu; Rong Chen; Bo Chen et al.	2025	Materials & Design	3	10.1016/j.matdes.2025.115222	
Comparison of intermediate-range order in GeO2 glass: Molecular dynamics using machine-learning interatomic potential vs reverse Monte Carlo fitting to experimental data	Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki	2024	The Journal of Chemical Physics	3	10.1063/5.0240087	
A Generalized Bond Switching Monte Carlo method for amorphous structure generation	Zhengneng Zheng; Fan Zheng; Yuning Wu et al.	2025	Computational Materials Today	3	10.1016/j.commt.2025.113002	4.18 (2026)
Atomic-scale factors that control the rate capability of nanostructured amorphous Si for high-energy-density batteries	Nongnuch Artrith; Alexander S. Urban; Yan Wang et al.	2019	arXiv (Cornell University)	3	10.48550/arxiv.1901.09272	
Machine learning metallic glass critical cooling rates through elemental and molecular simulation based featurization	Lane E. Schultz; Benjamin Afflerbach; Paul M. Voyles et al.	2024	Journal of Materiomics	3	10.1016/j.jmat.2024.100964	
When more data hurts: Optimizing data coverage while mitigating diversity-induced underfitting in an ultrafast machine-learned potential	Jason Gibson; Tesia D. Janicki; Ajinkya C. Hire et al.	2025	Physical Review Materials	3	10.1103/physrevmaterials.9.043802	
Insights into radiation damage in YBa2Cu3O7− from machine learned interatomic potentials	Ashley Dickson; Niccolò Di Eugenio; Federico Ledda et al.	2026	Superconductivity	2	10.1016/j.supcon.2026.100264	
Critical Composition Window for Co–P Bulk Metallic Glasses Revealed by Machine-Learned Interatomic Potentials	Md Sharif Khan; Bourgeois Biova Irénée Gadjagboui; Oliviero Andreussi	2025	The Journal of Physical Chemistry C	2	10.1021/acs.jpcc.5c05928	
The structural evolution of SiBCNZr amorphous ceramics analyzed by machine learning potential	Meng Zhang; Siyan Deng; Jianghong Zhang et al.	2024	Ceramics International	2	10.1016/j.ceramint.2024.[illegible]	[illegible] (2026)
Unraveling the Crystallization Kinetics of the Ge$_2$Sb$_2$Te$_5$ Phase Change Compound with a Machine-Learned Interatomic Potential	Omar Abou El Kheir; Luigi Bonati; Michele Parrinello et al.	2023	arXiv (Cornell University)	2	10.48550/arxiv.2304.03109	
Thermal Conductivity Modeling using Machine Learning Potentials: Application to Crystalline and Amorphous Silicon	Xin Qian; Shenyou Peng; Xiaobo Li et al.	2019	arXiv (Cornell University)	2	10.48550/arxiv.1907.09088	
Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics	Yanzhou Wang; Zheyong Fan; Ping Qian et al.	2024	Aaltodoc (Aalto University)	2	10.48550/arxiv.2408.[illegible]	[illegible] (2026)
Strong lattice anharmonicity and glass-like lattice thermal conductivity in nitrohalide double antiperovskites: A case study based on machine-learning potentials	Yuan Li; Weidong Zheng; Jin Huan Pu et al.	2025	Thermo-X	2	10.70401/tx.2025.0001	
Crystal-like thermal transport in amorphous carbon	Jaeyun Moon; Zhiting Tian	2024	arXiv (Cornell University)	2	10.48550/arxiv.2405.07298	
Universal Interatomic Potentials with DFT for Understanding Orbital Localization in Polydimethylsiloxane-Amorphous Silica Nanocomposites	Carson Farmer; Hector Medina	2025	ACS Omega	2	10.1021/acsomega.5c[illegible]	[illegible] (2026)

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Advanced active learning techniques for precision-driven machine-learned potentials in Sb2S3	V Priyadarshini; Pengfei Zou; Debashish Dash et al.	2025	Results in Engineering	2	10.1016/j.rineng.2025.106626	
Atomic armor for thermal stability in nanoporous structures	Rui Yang; Qiaoling Si; Qiang Sheng et al.	2025	Proceedings of the National Academy of Sciences	2	10.1073/pnas.2510746122	
Physics-informed machine learning exploration of Na storage mechanisms in disordered carbon	Nikhil Rampal; Stephen E. Weitzner; Fredrick Omenya et al.	2026	Energy storage materials	2	10.1016/j.ensm.202613.04867 (2026)	
Thermal conductivity of Li$_3$PS$_4$ solid electrolytes with ab initio accuracy	Davide Tisi; Federico Grasselli; Lorenzo Gigli et al.	2024	arXiv (Cornell University)	2	10.48550/arxiv.2401.12936	
Unveiling the crystallization kinetics in Ge-rich Ge$_x$Te alloys by large scale simulations with a machine-learned interatomic potential	Dario Baratella; Omar Abou El Kheir; Marco Bernasconi	2024	arXiv (Cornell University)	1	10.48550/arxiv.2404.15128	
On the boroxol ring fraction in melt-quenched B2O3 glass: Insights from machine learning potentials	Debendra Meher; Nikhil V. S. Avula; Sundaram Balasubramanian	2026	The Journal of Chemical Physics	1	10.1063/5.0321441	
Accurate prediction of structural and mechanical properties on amorphous materials enabled through machine-learning potentials: a case study of silicon nitride	Ganesh Kumar Nayak; Prashanth Srinivasan; Juraj Todt et al.	2024	DESY Publication Database (PUBDB) (Deutsches Elektronen-Synchrotron)	1	10.48550/arxiv.24080.0578(2026)	
Developing a Neural Network potential to investigate interface phenomena in solid-phase epitaxy	Ruggero Lot; Layla Martin-Samos; Stefano de Gironcoli et al.	2021		1	10.1109/nmdc50713.2021.9677541	
A Unified Deep Neural Network Potential Capable of Predicting Thermal Conductivity of Silicon in Different Phases	Ruiyang Li; Eungkyu Lee; Tengfei Luo	2019	arXiv (Cornell University)	1	10.48550/arxiv.1912.05044	
Simulation of structure and dynamics of phosphate glasses with fluoride additives: A hybrid approach combining universal and specialized neural network potentials	I. A. Balyakin; K. A. Arslanov; Maxim I. Vlasov	2025	Computational Materials Science	1	10.1016/j.commatsci.2025.114065 (2026)	
Nature of adsorption of amino acids and precursors on interstellar amorphous solid water	Natsuki Watanabe; Yuta Hori; Kazuki Okazawa et al.	2025	Monthly Notices of the Royal Astronomical Society	1	10.1093/mnras/staf1913	
AI-Aided Mapping of the Structure-Composition-Conductivity Relationships of Glass-Ceramic Lithium Thiophosphate Electrolytes	Haoyue Guo; Qian Wang; Alexander E. Urban et al.	2022	arXiv (Cornell University)	1	10.48550/arxiv.2201.11203	
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Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures	Zhuo Chen; Yuejin Yuan; Yanzhou Wang et al.	2025	arXiv (Cornell University)	1	10.48550/arxiv.2502.05584	
Atomistic Insights into Stress-Driven Lithiation at Silicon Anode Crack Tips	Bowen Zhang; Peiyao Zhang; Changguo Wang et al.	2025	ACS Applied Materials & Interfaces	1	10.1021/acsami.5c18372 (2026)	
Form and Function of Disorder	Parthapratim Biswas; Gang Chen; Serge Nakhmanson et al.	2021	physica status solidi (b)	1	10.1002/pssb.202100366	

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Nature of the amorphous-amorphous interfaces in solid-state batteries revealed using machine-learned interatomic potentials	Chuhong Wang; Muratahan Aykol; Tim Mueller	2023	ChemRxiv	0	10.26434/chemrxiv-2023-frr79	0.467 (2026)
Investigating Ionic Diffusivity in Amorphous Solid Electrolytes using Machine Learned Interatomic Potentials	Aqshat Seth; Rutvij Pankaj Kulkarni; Gopalakrishnan Sai Gautam	2024	arXiv (Cornell University)	0	10.48550/arxiv.2409.06242	
Machine-learned interatomic potentials for modelling nanoscale fracturing in silica and basalt	Marthe Grønlie Guren; Henrik Anderson Sveinsson; Anders Malthe-Sørenssen et al.	2023		0	10.5194/egusphere-egu23-6256	
Dynamic Metrics to Validate the Use of Machine-Learned Interatomic Potential for Dynamic Sampling	Bourgeois Biova Irénée Gadjagboui; Md Sharif Khan; Oliviero Andreussi	2026		0	10.1145/3773966.3785507	
Dependence of a cooling rate on structural and vibrational properties of amorphous silicon: A neural network potential-based molecular dynamics study	; Yasunobu Ando	2019	Institutional Repositories DataBase (IRDB)	0	(link)	
Critical Composition Window for Co–P Bulk Metallic Glasses Revealed by Machine-Learned Interatomic Potentials	Md Sharif Khan; Bourgeois Biova Irénée Gadjagboui; Oliviero Andreussi	2025	ChemRxiv	0	10.26434/chemrxiv-2025-vl9mj	0.467 (2026)
Suppression of Rayleigh Scattering in Silica Glass by Codoping Boron and Fluorine: Molecular Dynamics Simulations with ForceMatching and Neural Network Potentials	; Nobuhiro Nakamura; et al.	2022	Institutional Repositories DataBase (IRDB)	0	(link)	
Insights Into Radiation Damage in YBa$_2$Cu$_3$O$_{7-}$ From Machine-Learned Interatomic Potentials	Ashley Dicksona; Niccolò Di Eugenio; Federico Ledda et al.	2025	arXiv (Cornell University)	0	(link)	
Modeling Carbon Nanodomain Formation in Amorphous SiOC Using MACE Neural Network Potentials	Tuuli Tettenborn	2026	Digital Library of the University of Innsbruck (University of Innsbruck)	0	(link)	0.006 (2026)
Insights Into Radiation Damage in YBa$_2$Cu$_3$O$_{7-}$ From Machine-Learned Interatomic Potentials	Ashley Dicksona; Niccolò Di Eugenio; Federico Ledda et al.	2025	arXiv (Cornell University)	0	10.48550/arxiv.2512.24430	
Ion Dynamics in Amorphous Solid Electrolytes Studied Using Neural Network Potentials	Kōji Shimizu	2024	The Materials Research Society series	0	10.1007/978-981-97-6039-8_35	
On the Boroxol Ring Fraction in Melt-Quenched B2O3 Glass: Insights from Machine Learning Potentials	Nikhil Avula; Sundaram Balasubramanian; Debendra Meher	2026	AIP Publishing	0	10.60893/figshare.jcp.c.[illegible]831.v1	0.[illegible] (2026)
A neural network potential for the phase change material gete: large scale molecular dynamics simulations with close to ab initio accuracy	Gabriele C. Sosso	2013	BOA (University of Milano-Bicocca)	0	(link)	0.014 (2026)
On the Boroxol Ring Fraction in Melt-Quenched B2O3 Glass: Insights from Machine Learning Potentials	Nikhil Avula; Sundaram Balasubramanian; Debendra Meher	2026	AIP Publishing	0	10.60893/figshare.jcp.c.[illegible]831	0.[illegible] (2026)
Revealing Phonon Bridge Effect for Amorphous vs Crystalline Metal–Silicide Layers at Si/Ti Interfaces by a Machine Learning Potential	Mayur Pratap Singh; Lokanath Patra; Cheng Zhang et al.	2026	ACS Applied Electronic Materials	0	10.1021/acsaelm.5c0[illegible]	[illegible] (2026)
Investigation of medium-range order in amorphous GeTe and Ge2Sb2Te5 using density functional theory and neural network potential		2020	Seoul National University Open Repository (Seoul National University)	0	(link)	

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Charge integrated graph neural network-based machine learning potential for amorphous and non-stoichiometric hafnium oxide	Hyo Gyeong Shin; Seong Hun Kim; Eun Ho Kim et al.	2025	npj Computational Materials	0	10.1038/s41524-025-01864-3	
Application of Neural Network Potentials to Materials Science Analysis of Partial Crystallization in Glass Structures and Ion Conduction Mechanisms	Kōji Shimizu	2025	Vacuum and Surface Science	0	10.1380/vss.68.350	
Desorption dynamics of interstellar molecule on amorphous solid water investigated by machine learning potential-based PaCS-MD simulation	Watanabe Natsuki; Kästner Johannes; Yuta Hori et al.	2026	Institutional Repositories DataBase (IRDB)	0	(link)	
Desorption Dynamics of Interstellar Molecule on Amorphous Solid Water Investigated by Machine Learning Potential-based PaCS-MD Simulation	Yuta Hori; Natsuki Watanabe; Johannes Kästner et al.	2026	AIP Publishing	0	10.60893/figshare.jcp.c.8284920	[illegible]
Desorption dynamics of interstellar molecule on amorphous solid water investigated by machine learning potential-based PaCS-MD simulation	Natsuki Watanabe; Johannes Kästner; Yuta Hori et al.	2026	The Journal of Chemical Physics	0	10.1063/5.0325531	
Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses	Ryuki Kayano; Yaohiro Inagaki; Ryuta Matsubara et al.	2024	ChemRxiv	0	10.26434/chemrxiv-2024-hb50k-v2	0.467 (2026)
Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses	Ryuki Kayano; Yaohiro Inagaki; Ryuta Matsubara et al.	2024	ChemRxiv	0	10.26434/chemrxiv-2024-hb50k	0.467 (2026)
Neutron and X-ray Diffraction Reveal the Limits of Long-Range Machine Learning Potentials for Medium-Range Order in Silica Glass	Sai Harshit Balantrapu; Atul C. Thakur; Chris J. Benmore et al.	2026	arXiv (Cornell University)	0	(link)	
Neutron and X-ray Diffraction Reveal the Limits of Long-Range Machine Learning Potentials for Medium-Range Order in Silica Glass	Sai Harshit Balantrapu; Atul C. Thakur; Chris J. Benmore et al.	2026	arXiv (Cornell University)	0	10.48550/arxiv.2604.21222	
Neutron and x-ray diffraction reveal the limits of long-range machine learning potentials for medium-range order in silica glass	Sai Harshit; Atul C. Thakur; Chris J. Benmore et al.	2026	Journal of Physics Materials	0	10.1088/2515-7639/ae8643	
A systematic approach to generating accurate neural network potentials: the case of carbon	Yusuf Shaidu; Emine Küçükbenli; Ruggero Lot et al.	2021	npj Computational Materials	0	10.1038/s41524-021-00508-6	
High-dimensional neural-network potentials for phase change materials for data storage	Marco Bernasconi	2012	BOA (University of Milano-Bicocca)	0	(link)	0.014 (2026)
ALDo 2.0: Scalable thermal transport from first principles and machine learning potentials	Giuseppe Barbalinardo; Zekun Chen; Dylan A. Folkner et al.	2026	Mendeley Data	0	10.17632/t3f42nv4fw.1	
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Dynamic heterogeneity in sodium silicate melts via machine-learning potential	Kumpei Shiraishi; Rikuta Nozawa; Emi Minamitani	2026	arXiv (Cornell University)	0	(link)	
Dynamic heterogeneity in sodium silicate melts via machine-learning potential	Kumpei Shiraishi; Rikuta Nozawa; Emi Minamitani	2026	arXiv (Cornell University)	0	10.48550/arxiv.2606.26646	
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Density functional and neural-network potential simulations of Ag migration in disordered GeS 2 electrolytes	Jaakko Akola; Sondre Dahl; A Götz et al.	2026	Physical Review Materials	0	10.1103/75rr-pqqp	
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Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses using Machine Learning Potential	Shingo Urata	2022	arXiv (Cornell University)	0	10.48550/arxiv.2209.00316	
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Study of vacancy ordering and the boson peak in metastable cubic Ge-Sb-Te using machine learning potentials	Young-Jae Choi; Minjae Ghim; Seung-Hoon Jhi	2023	arXiv (Cornell University)	0	10.48550/arxiv.2309.01089	
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Machine Learning Guided Exploration of Amorphous Electrodes and Electrolytes	Sai Gautam Gopalakrishnan; Aqshat Seth; Debsundar Dey et al.	2025	ECS Meeting Abstracts	0	10.1149/ma2025-0271034mtgabs	0.091 (2026)
Large-scale first-principle simulations of amorphous indium oxide	Matthew Bousquet; Francois Gygi; Giulia Galli	2026	arXiv (Cornell University)	0	(link)	
Large-scale first-principle simulations of amorphous indium oxide	Matthew Bousquet; Francois Gygi; Giulia Galli	2026	arXiv (Cornell University)	0	10.48550/arxiv.2607.08617	
Amorphous boron nitride as an ultrathin copper diffusion barrier for advanced interconnects	Onurcan Kaya; Hyeongjoon Kim; Byeongkyu Kim et al.	2025	2D Materials	0	10.1088/2053-1583/ae2521	4.053 (2026)
Large scale molecular modelling of basalt surfaces and fracturing in basaltic glass	Marthe Grønlie Guren; Henrik Andersen Sveinsson; Razvan Caracas et al.	2024		0	10.5194/egusphere-egu24-11861	

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Large scale simulations of amorphous phase change materials for data storage	Marco Bernasconi	2012	BOA (University of Milano-Bicocca)	0	(link)	0.014 (2026)
The ab initio amorphous materials database: Empowering machine learning to decode diffusivity	Hui Zheng; Patrick Huck	2024	OSTI OAI (U.S. Department of Energy Office of Scientific and Technical Information)	0	10.17188/mpcontribs/2281590	
Energy dissipation at the atomic scale explains how fracture energy depends on crack velocity in silica glass	Marthe Grønlie Guren; Sigbjørn Løland Bore; François Renard et al.	2026	arXiv (Cornell University)	0	10.48550/arxiv.2605.03457	
Energy dissipation at the atomic scale explains how fracture energy depends on crack velocity in silica glass	Marthe Grønlie Guren; Sigbjørn Løland Bore; François Renard et al.	2026	arXiv (Cornell University)	0	(link)	
Role of Characteristic Length Scale in Interface Graphitization-Induced Wear Resistance of Diamond and Amorphous Carbon	Wenliang Shi; YanYu Zhang; WenZheng Liao et al.	2026	arXiv (Cornell University)	0	(link)	
Role of Characteristic Length Scale in Interface Graphitization-Induced Wear Resistance of Diamond and Amorphous Carbon	Wenliang Shi; YanYu Zhang; WenZheng Liao et al.	2026	arXiv (Cornell University)	0	10.48550/arxiv.2606.03325	
Neural Network-Based Simulation Method to Examine Ion Behaviors under Electric Fields: Application to Ion Migration in Amorphous Li3PO4	Kōji Shimizu; Ryuji Otsuka; Satoshi Watanabe	2023	Journal of the Japan Society of Powder and Powder Metallurgy	0	10.2497/jjspm.23-00043	
Machine-Learned Potential–Driven Insights into Li S–P S Glass Electrolytes and Li PS Cl Interfacial Chemistry for Solid-State Batteries	Byungju Lee; C S Kim; Sojeong Yang	2026	ECS Meeting Abstracts	0	10.1149/ma2026-016700mtgabs	0.091 (2026)
Suppressed glass transition behavior through high configurational entropy in high-entropy metallic glasses	Jinyue Wang; Fuzhi Dai; Yihuan Cao et al.	2026	Acta Materialia	0	10.1016/j.actamat.2026.122301	1.712 (2026)
Visualizing a Li-depleted amorphous cathode–electrolyte interphase in sulfide solid-state batteries using in situ cryogenic electron microscopy	Yuki Nomura; Ryoma Sasaki; Huu Duc Luong et al.	2026	ChemRxiv	0	10.26434/chemrxiv.15003253/v1	0.462 (2026)
Quantitative assessment of the structure of (Na2O)0.1-(V2O5)0.6-(TeO2)0.3 glass: An ab initio and machine learning interatomic potential study of oxide systems with mixed-valence	Firas Shuaib; Rémi Piotrowski; Gaëlle Delaizir et al.	2026	Artificial Intelligence Chemistry	0	10.1016/j.aichem.2026.100103	9.342 (2026)
Thermochemical properties and glass transition of Na 2 O–Al 2 O 3 –P 2 O 5 system: a deep learning potential approach and its experimental validation	Dmitry Zakiryanov; Maxim I. Vlasov; Victor Bykov	2026	Modelling and Simulation in Materials Science and Engineering	0	10.1088/1361-651x/ae39f7	
Study on Cu 50 Zr 50 Metallic Glass Based on Machine Learning Force Field: Correlation between Virial and Equipartition Theorem Violations and Atomic-Scale Dynamic Behaviors	Wenliang Shi; YanYu Zhang; HongYu Wu et al.	2026	Chinese Physics B	0	10.1088/1674-1056/ae42ba	1.149 (2026)
Zero-Dimensional Ice Ring Topological Proton Glass: Gapless Texture Soft Modes, Anharmonic Wandering, and Three Conserved Invariants	Zhongzheng Miao; Miao Zhongzheng	2026	ChemRxiv	0	10.26434/chemrxiv.15004427/v1	0.463 (2026)

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Nanoparticules d'argent enrobées dans une matrice de silice pour des dispositifs antibactériens : synthèse en phase gazeuse et modélisation assistée par apprentissage automatique	Marie-Salomé Trillot	2025	HAL (Le Centre pour la Communication Scientifique Directe)	0	(link)	0.03 (2026)
Influence of crystal orientation and polymorphism on the shock response of boron carbide	Kimia Ghaffari; Salil Bavdekar; Douglas E. Spearot et al.	2025	Acta Materialia	0	10.1016/j.actamat.2025.121305	10.172 (2026)
Phonon Transport in Disordered Two-Dimensional Graphene	Hassan Shoaib	2025	McGill-DEV	0	10.82308/27920	
Atomic structure at the surface of warm basaltic glasses	Marthe Grønlie Guren; Henrik Anderson Sveinsson; Razvan Caracas et al.	2025		0	10.5194/egusphere-egu25-12944	
Through Rainbow-Tinted Glasses: Machine Learning-Driven Modeling of Chromophores	Eric Lindgren	2026		0	10.63959/chalmers.dt/5890	
When More Data Hurts: Optimizing Data Coverage While Mitigating Diversity Induced Underfitting in an Ultra-Fast Machine-Learned Potential	Jason B. Gibson; Tesia Janicki; Ajinkya C. Hire et al.	2024	arXiv (Cornell University)	0	10.48550/arxiv.2409.07610	
Dynamic Anion Sublattices, Cooperative Ion Migration and Metastable Pathway Engineering for Next-Generation Solid Electrolytes	Rajab Abbas; Rabia Shahzad; Aasma Bibi et al.	2026	Saudi Journal of Engineering and Technology	0	10.36348/sjet.2026.v11i08.003	
Structural origin of disorder-induced ion conduction in NaFePO4 cathode materials	Rasmus Christensen; Kristin A. Persson; Morten M. Smedskjær	2025	ChemRxiv	0	10.26434/chemrxiv-2025-9sf4n	0.467 (2026)
MOLECULAR SIMULATION METHODS FOR MODELING	Dylan David Fortney	2026	Purdue	0	10.25394/pgs.32048331.v1	
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Nuclear Quantum Effects in the Acetylene:Ammonia Plastic Co-crystal	Atul C. Thakur; Richard C. Remsing	2023	arXiv (Cornell University)	0	10.48550/arxiv.2310.00480	
Beyond Boltzmann transport: Green–Kubo prediction of lattice thermal conductivity with machine-learned potentials	Xiaoying Zhang; Ning Liu; Qian Guo et al.	2026	The Journal of Chemical Physics	0	10.1063/5.0297462	
Revisit Many-body Interaction Heat Current and Thermal Conductivity Calculation in Moment Tensor Potential/LAMMPS Interface	Tai, Siu Ting; Chen Wang; Ruihuan Cheng et al.	2024	arXiv (Cornell University)	0	10.48550/arxiv.2411.01255	
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Thermal Transport Mechanisms of Phonons, Diffusons, Electrons, and Ions in Functional Materials	Yatian Zhang	2026	Media (https://www.suub.uni-bremen.de/)	0	10.26092/elib/5520	
Accelerating Amorphous Alloy Discovery: Data-Driven Property Prediction via General-Purpose Machine Learning Interatomic Potential	Xuhe Gong; Hengbo Zhao; Xiao Fu et al.	2025	arXiv preprint	0	(link)	
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Persistent homology-based descriptor for machine-learning potential of amorphous structures	Emi Minamitani; Ippei Obayashi; Koji Shimizu et al.	2022	arXiv preprint	0	(link)	
Unveiling the crystallization kinetics in Ge-rich Ge$_x$Te alloys by large scale simulations with a machine-learned interatomic potential	Dario Baratella; Omar Abou El Kheir; Marco Bernasconi	2024	arXiv preprint	0	(link)	
Melt-Quench Failures and Practical Solutions for Universal Machine-Learning Interatomic Potentials in Amorphous Structure Generation	Shuwei Li; Yuqi An; Xingyu Guo et al.	2026	arXiv preprint	0	(link)	
Comparison of intermediate-range order in GeO$_2$ glass: molecular dynamics using machine-learning interatomic potential vs.\ reverse Monte Carlo fitting to experimental data	Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki	2024	arXiv preprint	0	(link)	
Machine-learned potential for amorphous Indium-Tin-Oxide alloys	Shuaiyang Guo; Yuan Wang; Wei Zhang	2026	arXiv preprint	0	(link)	
Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses using Machine Learning Potential	Shingo Urata	2022	arXiv preprint	0	(link)	
Large-scale atomistic study of plasticity in amorphous gallium oxide with a machine-learning potential	Jiahui Zhang; Junlei Zhao; Jesper Byggmästar et al.	2024	arXiv preprint	0	(link)	
Validation Workflow for Machine Learning Interatomic Potentials for Complex Ceramics	Kimia Ghaffari; Salil Bavdekar; Douglas E. Spearot et al.	2024	arXiv preprint	0	10.1016/j.commatsci.2024.112983	
Atomistic Insights into Cu/amorphous-Ta$_x$N Interfacial Adhesion via Machine Learning Interatomic Potentials: Effects of Stoichiometry and Interface Construction	Jeong Min Choi; Jaehoon Kim; Ji-Hwan Lee et al.	2025	arXiv preprint	0	(link)	
Machine Learning Potential Powered Insights into the Mechanical Stability of Amorphous Li-Si Alloys	Zixiong Wei; Nongnuch Artrith	2024	arXiv preprint	0	(link)	
Is the Future of Materials Amorphous? Challenges and Opportunities in Simulations of Amorphous Materials	Ata Madanchi; Emna Azek; Karim Zongo et al.	2024	arXiv preprint	0	(link)	
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Resolving Structural Avalanches in Amorphous Carbon with Arclength Continuation	Fraser Birks; Ibrahim Ghanem; Lars Pastewka et al.	2026	arXiv preprint	0	10.1103/6n5m-rxc1	
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Thermal Conductivity and Temperature-Induced Band Gap Renormalization in Crystalline and Amorphous Ga$_2$O$_3$	Rustam Arabov; Jiaxuan Li; Xiaotong Chen et al.	2026	arXiv preprint	0	(link)	
From oligomers to entangled polymers: How to train a transferable machine learning interatomic potential	Mirko Fischer; Andreas Heuer	2026	arXiv preprint	0	(link)	

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Polymorphic crystallites model for monolayer amorphous materials	Le-Ye Zhu; Xi Zhang; Yun-Peng Wang et al.	2026	arXiv preprint	0	(link)	
A Unified Deep Neural Network Potential Capable of Predicting Thermal Conductivity of Silicon in Different Phases	Ruiyang Li; Eungkyu Lee; Tengfei Luo	2019	arXiv preprint	0	(link)	
Atomistic insights into hydrogen migration in IGZO from machine-learning interatomic potential: linking atomic diffusion to device performance	Hyunsung Cho; Minseok Moon; Jaehoon Kim et al.	2025	arXiv preprint	0	(link)	
Thermal Conductivity Modeling using Machine Learning Potentials: Application to Crystalline and Amorphous Silicon	Xin Qian; Shenyou Peng; Xiaobo Li et al.	2019	arXiv preprint	0	(link)	
Amorphous materials as a frontier challenge for universal interatomic potentials	Natascia L. Fragapane; Volker L. Deringer	2026	arXiv preprint	0	(link)	
Efficient training of machine learning potentials for metallic glasses: CuZrAl validation	Antoni Wadowski; Anshul D. S. Parmar; Filip Kaśkosz et al.	2024	arXiv preprint	0	(link)	
Constitutive relations for plasticity of amorphous carbon	Richard Jana; Julian von Lautz; S. Mostafa Khosrownejad et al.	2020	J. Phys. Mater. 3, 035005 (2020)	0	10.1088/2515-7639/ab953c	
Crystal-like thermal transport in amorphous carbon	Jaeyun Moon; Zhiting Tian	2024	arXiv preprint	0	(link)	
Structural and elastic properties of amorphous carbon from simulated quenching at low rates	Richard Jana; Daniele Savio; Volker L. Deringer et al.	2019	Modelling Simul. Mater. Sci. Eng. 27, 085009 (2019)	0	10.1088/1361-651X/ab45da	
A Gaussian Approximation Potential for Amorphous Si:H	Davis Unruh; Reza Vatan Meidanshahi; Stephen M. Goodnick et al.	2021	arXiv preprint	0	(link)	
On the origin of in-gap states in amorphous Ge$_2$Sb$_2$Te$_5$	Omar Abou El Kheir; Marco Bernasconi	2026	arXiv preprint	0	(link)	
A Systematic Approach to Generating Accurate Neural Network Potentials: the Case of Carbon	Yusuf Shaidu; Emine Kucukbenli; Ruggero Lot et al.	2020	arXiv preprint	0	(link)	
Modulation of structural short-range order due to chemical patterning in multi-component amorphous interfacial complexions	Esther C. Hessong; Zhengyu Zhang; Tianjiao Lei et al.	2025	arXiv preprint	0	(link)	
Medium-range structural order in amorphous arsenic	Yuanbin Liu; Yuxing Zhou; Richard Ademuwagun et al.	2025	arXiv preprint	0	(link)	
From Bulk to Surface: Structure and Dynamics of Amorphous Alumina from Deep Potential Molecular Dynamics	Zheng Yu; Jiayan Xu; Abhirup Patra et al.	2026	arXiv preprint	0	(link)	
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Bridging classical and quantum interpretation of chemical state analysis by XPS/HAXPES to resolve short-range order in amorphous alumina films	Simon Gramatte; Xing Wang; Michael Alejandro Hernández Bertrán et al.	2024	arXiv preprint	0	(link)	
Machine-Learning-Guided Insights into Solid-Electrolyte Interphase Conductivity: Are Amorphous Lithium Fluorophosphates the Key?	Peichen Zhong; Kristin A. Persson	2025	arXiv preprint	0	10.1021/acsenergylett.5c03526	
Atomic insight into Li$\^+$ ion transport in amorphous electrolytes Li$_x$AlO$_y$Cl$_{3+x-2y}$ (0.5 $\leq$ x $\leq$ 1.5, 0.25 $\leq$ y $\leq$ 0.75)	Yang Qifan; Xu Jing; Fu Xiao et al.	2024	arXiv preprint	0	(link)	

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Realistic atomistic structure of amorphous silicon from machine-learning-driven molecular dynamics	Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.	2018	arXiv preprint	0	(link)	
Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics	Yanzhou Wang; Zheyong Fan; Ping Qian et al.	2024	arXiv preprint	0	(link)	
The ab initio amorphous materials database: Empowering machine learning to decode diffusivity	Hui Zheng; Eric Sivonxay; Max Gallant et al.	2024	arXiv preprint	0	(link)	
Structural properties of amorphous Na$_3$OCl electrolyte by first-principles and machine learning molecular dynamics	T. -L. Pham; M. Guerboub; S. D. Wansi Wendj et al.	2024	arXiv preprint	0	(link)	
Enhanced ionic conductivity through crystallization of glass-Li$_3$PS$_4$ by machine learning molecular dynamics simulations	Koji Shimizu; Parth Bahuguna; Shigeo Mori et al.	2023	arXiv preprint	0	(link)	
Machine learning potential for predicting thermal conductivity of -phase and amorphous Tantalum Nitride	Zhicheng Zong; Yangjun Qin; Jiahong Zhan et al.	2025	arXiv preprint	0	(link)	
Optimization and Validation of a Deep Learning CuZr Atomistic Potential: Robust Applications for Crystalline and Amorphous Phases with near-DFT Accuracy	Christopher M. Andolina; Philip Williamson; Wissam A. Saidi	2020	J. Chem. Phys. 152, 154701 (2020)	0	10.1063/5.0005347	
Mechanisms of alkali ionic transport in amorphous oxyhalides solid state conductors	Luca Binci; KyuJung Jun; Bowen Deng et al.	2026	arXiv preprint	0	(link)	
Neural-Network Assisted Study of Nitrogen Atom Dynamics on Amorphous Solid Water – II. Diffusion	Viktor Zaverkin; Germán Molpeceres; Johannes Kästner	2021	arXiv preprint	0	10.1093/mnras/stab3631	
Machine Learning Assisted Modeling of Amorphous TiO$_2$-Doped GeO$_2$ for Advanced LIGO Mirror Coatings	Jun Jiang; Rui Zhang; Kiran Prasai et al.	2025	arXiv preprint	0	(link)	
Fast Isotropic Li-Ion Diffusion in Zeolitic Imidazolate Framework Glass Electrolytes for Batteries	Yong Li; Tao Du; Timothée Jamin et al.	2026	arXiv preprint	0	(link)	
kALDo 2.0: Scalable Thermal Transport from First Principles and Machine Learning Potentials	Giuseppe Barbalinardo; Zekun Chen; Dylan Folkner et al.	2026	arXiv preprint	0	(link)	
Neural-Network Assisted Study of Nitrogen Atom Dynamics on Amorphous Solid Water. I. Adsorption & Desorption	Germán Molpeceres; Viktor Zaverkin; Johannes Kästner	2020	arXiv preprint	0	10.1093/mnras/staa2891	
Simulations of High Temperature Decomposition of Metal-Organic Frameworks to form Amorphous Catalysts	Connor W. Edwards; Oliver M. Linder-Patton; Jack D. Evans	2026	arXiv preprint	0	(link)	
Thermal conductivity of glasses above the plateau: first-principles theory and applications	Michele Simoncelli; Francesco Mauri; Nicola Marzari	2022	arXiv preprint	0	(link)	
Simulating alternating bias assisted annealing of amorphous oxide tunnel junctions	Alexander C. Tyner; Alexander V. Balatsky	2025	arXiv preprint	0	(link)	
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Thermal transport of glasses via machine learning driven simulations	Paolo Pegolo; Federico Grasselli	2024	arXiv preprint	0	(link)	
Machine learning Hamiltonian enables scalable and accurate defect calculations: The case of oxygen vacancies in amorphous SiO$_2$	Zhenxing Dai; Zhong Yang; Mingjue Ni et al.	2026	arXiv preprint	0	(link)	
Synthesizability, hardness, and stacking order in multicomponent transition metal carbides from machine-learned potentials	Xin Liu; Anirudh Raju Natarajan	2026	arXiv preprint	0	(link)	
Efficient and accurate simulation of vitrification in multi-component metallic liquids with neural-network potentials	Rui Su; Jieyi Yu; Pengfei Guan et al.	2022	arXiv preprint	0	(link)	
Atomic cluster expansion potential for the Si-H system	Louise A. M. Rosset; Volker L. Deringer	2025	arXiv preprint	0	(link)	
Mechanisms of temperature-dependent thermal transport in amorphous silica from machine-learning molecular dynamics	Ting Liang; Penghua Ying; Ke Xu et al.	2023	Physical Review B 108, 184203 (2023)	0	10.1103/PhysRevB.108.184203	
Atomic-scale factors that control the rate capability of nanostructured amorphous Si for high-energy-density batteries	Nongnuch Artrith; Alexander Urban; Yan Wang et al.	2019	arXiv preprint	0	(link)	
CO Adsorption Sites on Interstellar Water Ices Explored with Machine Learning Potentials. Binding energy distributions and snowline	Giulia M. Bovolenta; Germán Molpeceres; Kenji Furuya et al.	2025	A&A 703, A172 (2025)	0	10.1051/0004-6361/202555836	
AI-Aided Mapping of the Structure-Composition-Conductivity Relationships of Glass-Ceramic Lithium Thiophosphate Electrolytes	Haoyue Guo; Qian Wang; Alexander Urban et al.	2022	arXiv preprint	0	(link)	
Nine-element machine-learned interatomic potentials for multiphase refractory alloys	Jesper Byggmästar; Tiago Lopes; Zheyong Fan et al.	2026	arXiv preprint	0	(link)	
Reaction dynamics on amorphous solid water surfaces using interatomic machine learned potentials. Microscopic energy partition revealed from the P + H -> PH reaction	Germán Molpeceres; Viktor Zaverkin; Kenji Furuya et al.	2023	A&A 673, A51 (2023)	0	10.1051/0004-6361/202346073	
Experiment-driven atomistic materials modeling: A case study combining X-ray photoelectron spectroscopy and machine learning potentials to infer the structure of oxygen-rich amorphous carbon	Tigany Zarrouk; Rina Ibragimova; Albert P. Bartók et al.	2024	J. Am. Chem. Soc. 146, 14645 (2024)	0	(link)	
When More Data Hurts: Optimizing Data Coverage While Mitigating Diversity Induced Underfitting in an Ultra-Fast Machine-Learned Potential	Jason B. Gibson; Tesia D. Janicki; Ajinkya C. Hire et al.	2024	arXiv preprint	0	(link)	
Large Scale Structure Prediction of Near-Stoichiometric Magnesium Oxide Based on a Machine-Learned Interatomic Potential: Novel Crystalline Phases and Oxygen-Vacancy Ordering	Hossein Tahmasbi; Stefan Goedecker; S. Alireza Ghasemi	2021	Phys. Rev. Materials 5, 083806 (2021)	0	10.1103/PhysRevMaterials.5.083806	

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Unveiling hole-facilitated amorphisation in pressure-induced phase transformation of silicon	Tong Zhao; Shulin Zhong; Yuxin Sun et al.	2024	arXiv preprint	0	(link)	
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Modeling phase separation in polymer-derived silicon carbonitride ceramics through extended machine learning molecular dynamics	Fabien Mortier; Sylvian Cadars; Olivier Masson et al.	2026	arXiv preprint	0	(link)	
Understanding phase transitions of $ $-quartz under dynamic compression conditions by machine-learning driven atomistic simulations	Linus C. Erhard; Christoph Otzen; Jochen Rohrer et al.	2024	npj Comput Mater 11, 58 (2025)	0	10.1038/s41524-025-01542-4	
Simulation of the crystallization kinetics of Ge$_2$Sb$_2$Te$_5$ nanoconfined in superlattice geometries for phase change memories	Debdipto Acharya; Omar Abou El Kheir; Simone Marcorini et al.	2025	arXiv preprint	0	(link)	
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How to Train a Shallow Ensemble	Moritz Schäfer; Matthias Kellner; Johannes Kästner et al.	2026	arXiv preprint	0	(link)	
Atomistic Structure Generation and Neural-Network Screening of Hard Carbons to Identify High-Capacity Sodium Storage	Harry Mclean; Aiden Daniel Emery; Theodore Thomas Walton et al.	2026	arXiv preprint	0	(link)	
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Million-atom simulation of the set process in phase change memories at the real device scale	Omar Abou El Kheir; Marco Bernasconi	2025	arXiv preprint	0	(link)	
Cluster expansion constructed over Jacobi-Legendre polynomials for accurate force fields	Michelangelo Domina; Urvesh Patil; Matteo Cobelli et al.	2022	arXiv preprint	0	(link)	
Nucleation mechanism for the direct graphite-to-diamond phase transition	Rustam Z. Khaliullin; Hagai Eshet; Thomas D. Kuhne et al.	2011	arXiv preprint	0	10.1038/nmat3078	
Structure and solidification of the (Fe0.75B0.15Si0.1)100-xTax (x=0-2) melts: experiment and machine learning	I. V. Sterkhova; L. V. Kamaeva; V. I. Ladyanov et al.	2022	arXiv preprint	0	10.1016/j.jpcs.2022.111143	
An experimentally validated end-to-end framework for operando modeling of intrinsically complex metallosilicates	Jong Hyun Jung; Tom Schächtel; Yongliang Ou et al.	2025	arXiv preprint	0	(link)	
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Sub-micrometer phonon mean free paths in metal-organic frameworks revealed by machine-learning molecular dynamics simulations	Penghua Ying; Ting Liang; Ke Xu et al.	2023	ACS Appl.Mater.Interfaces, 2023,15, 36412	0	10.1021/acsami.3c07770	

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Heat transport in superionic materials via machine-learned molecular dynamics	Wenjiang Zhou; Benrui Tang; Zheyong Fan et al.	2025	arXiv preprint	0	(link)	
Machine learning for predicting ultralow thermal conductivity and high ZT in complex thermoelectric materials	Yuzhou Hao; Yuting Zuo; Jiongzhi Zheng et al.	2024	arXiv preprint	0	10.1021/acsami.4c09043	
Incipient ionic conductors: Ion-constrained lattices achieving superionic-like thermal conductivity by extreme anharmonicity	Yongheng Li; Chunqiu Lu; Bin Wei et al.	2025	arXiv preprint	0	10.1002/adma.202513381	
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Delocalization of a disordered bosonic system by repulsive interactions	B. Deissler; Matteo Zaccanti; G. Roati et al.	2010	Nature Physics	273	10.1038/nphys1635	
Topological triple phase transition in non-Hermitian Floquet quasicrystals	Sebastian Weidemann; Mark Kremer; Stefano Longhi et al.	2022	Nature	225	10.1038/s41586-021-04253-0	
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Topological Phase Transitions and Mobility Edges in Non-Hermitian Quasicrystals	Quan Lin; Tianyu Li; Lei Xiao et al.	2022	Physical Review Letters	136	10.1103/physrevlett.129.113601	
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Bose-Einstein condensates in optical quasicrystal lattices	Laurent Sanchez-Palencia; L. Santos	2005	Physical Review A	117	10.1103/physreva.72.053607	
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Power dependent soliton location and stability in complex photonic structures	Yannis Kominis; Kyriakos Hizanidis	2008	Optics Express	73	10.1364/oe.16.012124	
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Photonic quasicrystal single-cell cavity mode	Sun-Kyung Kim; Jeehye Lee; Se-Heon Kim et al.	2005	Applied Physics Letters	45	10.1063/1.1852716	3.881 (2026)
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Observation of Topological Corner State Arrays in Photonic Quasicrystals	Aoqian Shi; Yiwei Peng; Jiapei Jiang et al.	2024	Laser & Photonics Review	35	10.1002/lpor.202300956	
Dephasing-Induced Mobility Edges in Quasicrystals	Stefano Longhi	2024	Physical Review Letters	34	10.1103/physrevlett.132.236301	
Enhanced Soliton Transport in Quasiperiodic Lattices with Introduced Aperiodicity	Andrey A. Sukhorukov	2006	Physical Review Letters	33	10.1103/physrevlett.96.113902	
Exact mobility edges and topological phase transition in two-dimensional non-Hermitian quasicrystals	Zhihao Xu; X. C. Xia; Shu Chen	2021	Science China Physics Mechanics and Astronomy	31	10.1007/s11433-021-1802-4	
Quasiperiodic photonic crystal fiber [Invited]	Exian Liu; Jianjun Liu	2023	Chinese Optics Letters	30	10.3788/col202321.060503	4.254 (2026)
Luminescence properties of a Fibonacci photonic quasicrystal	V. Passias; N. V. Valappil; Z. Shi et al.	2009	Optics Express	28	10.1364/oe.17.006636	
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Simultaneous large band gaps and localization of electromagnetic and elastic waves in defect-free quasicrystals	Tianbao Yu; Zhong Wang; Wenxing Liu et al.	2016	Optics Express	27	10.1364/oe.24.007951	
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Photonic band gaps and localization in two-dimensional metallic quasicrystals	Mehmet Bayındır; Ertugrul Cubukcu; İrfan Bulu et al.	2001	Europhysics Letters (EPL)	25	10.1209/epl/i2001-00485-9	2.066 (2026)
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Spin waves in planar quasicrystal of Penrose tiling	Justyna Rychły-Gruszecka; Szymon Mieszczak; Jarosław W. Kłos	2017	Journal of Magnetism and Magnetic Materials	20	10.1016/j.jmmm.2017.03.029	
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Topological wave phenomena in photonic time quasicrystals	Xiang Ni; Shixiong Yin; Huanan Li et al.	2025	Physical review. B./Physical review. B	8	10.1103/physrevb.111.125421	
Tunable defect modes through the (YBCO-Yttria) based on Octonacci photonic quasicrystals	Youssef Trabelsi; Francis Segovia-Chaves; Naim Ben Ali	2022	Results in Physics	8	10.1016/j.rinp.2022.106176	
Spectral properties of two coupled Fibonacci chains	Anouar Moustaj; Malte Röntgen; Christian V. Morfonios et al.	2023	New Journal of Physics	8	10.1088/1367-2630/acf0e0	
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Stability of quasicrystalline ultracold fermions to dipolar interactions	Paolo Molignini	2025	Physical Review Research	6	10.1103/szdc-61nl	
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Observation of Topological Corner State Arrays in Photonic Quasicrystals	Aoqian Shi; Yiwei Peng; Jiapei Jiang et al.	2022	arXiv (Cornell University)	5	10.48550/arxiv.2209.05751	

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Optical properties of 1D photonic time quasicrystals	Anny Araújo; Carlos Costa; S. Azevedo et al.	2025	Journal of the Optical Society of America B	4	10.1364/josab.542001	
Symmetry-Induced Light Confinement in a Photonic Quasicrystal-Based Mirrorless Cavity	Gianluigi Zito; Giulia Rusciano; Antonio Sasso et al.	2016	Crystals	4	10.3390/cryst6090113	1.465 (2026)
Resonant modes of 12-fold symmetric defect free photonic quasicrystal	Minfeng Chen; Yunjing Li; Yuh-Jen Cheng et al.	2014	Optics Express	4	10.1364/oe.22.002007	
Tunability in the terahertz range of a one-dimensional photonic quasicrystal containing an InSb semiconductor	Francis Segovia-Chaves; Hussein A. Elsayed; Ahmed Mehaney	2021	Optik	4	10.1016/j.ijleo.2021.167675	
Diffraction propagation of light wave and image distortionless transmission in ten-fold symmetric photonic quasicrystal lattices	Wentao Jin; Meng Song; Wen-Cheng Hu et al.	2021	Optics Communications	4	10.1016/j.optcom.2021.127329	
Properties of defect modes in two-dimensional organic octagonal quasicrystals at low dielectric contrast	Yuanyuan Cai; Zhi Wang; Xiao Chen et al.	2018	Optics Communications	4	10.1016/j.optcom.2018.06.010	
Isotropic photonic bandgap in Penrose textured metallic mircocavity	J. Y. Zhang; Hoi Lam Tam; W.H. Wong et al.	2006	Solid State Communications	4	10.1016/j.ssc.2006.02.035	
Anomalous optical Anderson localization in mixed one dimensional photonic quasicrystals	Cunding Liu; Mingdong Kong; Bincheng Li	2015	Optics Express	3	10.1364/oe.23.0a1297	
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Programmable 2D photonic quasicrystals with real-time reconfigurability	Peisheng Sun; Xuesong Hu; Ruonan Li et al.	2025	Optics Letters	3	10.1364/ol.568662	
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Properties of microcavities in two-dimensional photonic quasicrystals with octagonal rotational symmetry	D. M. Beggs; M. A. Kaliteevski; R. A. Abram	2007	Journal of Modern Optics	3	10.1080/09500340601102441	
Simultaneous localization of photons and phonons in defect-free dodecagonal phoxonic quasicrystals	Bihang Xu; Zhong Wang; Yixiang Tan et al.	2018	Modern Physics Letters B	3	10.1142/s0217984918500963	
Interface states in two-dimensional quasicrystals with broken inversion symmetry	Danilo Beli; Matheus I. N. Rosa; Luca Lomazzi et al.	2023	arXiv (Cornell University)	3	10.48550/arxiv.2304.09628	
Observation of Topological Corner State Arrays in Photonic Quasicrystals (Laser Photonics Rev. 18(7)/2024)	Aoqian Shi; Yiwei Peng; Jiapei Jiang et al.	2024	Laser & Photonics Review	2	10.1002/lpor.202470041	
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Light wave propagation and diffraction inhibition in three-dimensional photonic quasicrystal lattices	Meng Song; Jin Wentao; Shaochun Fu et al.	2023	Results in Physics	2	10.1016/j.rinp.2023.106884	
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Topological states in Penrose-square photonic crystals	Qichen Zhang; Jianzhi Chen; Dongyang Liu et al.	2024	Journal of the Optical Society of America A	2	10.1364/josaa.520606	
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Mode localization in defect-free GaN-based bottom-up photonic quasicrystal lasers	Tzeng-Tsong Wu; Chih-Cheng Chen; Hao Chen et al.	2012		1	10.1109/islc.2012.6348389	
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Formation of nonlinear modes in one-dimensional quasiperiodic lattices with a mobility edge	Dmitry A. Zezyulin; G. L. Alfimov	2024	Physical Review A	1	10.1103/physreva.110.063304	
Coupling photonic quasicrystals with a one-dimensional critical high-temperature superconducting cavity	Francis Segovia-Chaves; Youssef Trabelsi; T.A. Taha	2023	Optik	1	10.1016/j.ijleo.2023.170847	
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Vortex solitons in disclination quasicrystals	Hua Zhong; Yaroslav Kartashov; Yongdong Li et al.	2026	Reports on Progress in Physics	1	10.1088/1361-6633/ae73b5	
A tribute to Sydney G. Davison	Aart W. Kleyn	2003	Progress in Surface Science	1	10.1016/j.progsurf.2003.09.003	
Stability of quasicrystalline ultracold fermions to dipolar interactions	Paolo Molignini	2024	arXiv (Cornell University)	1	10.48550/arxiv.2403.04830	
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Photonic band gap effect and localization in two-dimensional Penrose lattice	Mehmet Bayındır; Ertugrul Cubukcu; İrfan Bulu et al.	2002		0	10.1109/qels.2001.961940	
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Critical states and anomalous wave transport in an aperiodic polariton monotile	Valtýr Kári Daníelsson; Helgi Sigurðsson	2026	arXiv (Cornell University)	0	10.48550/arxiv.2605.29023	
Critical states and anomalous wave transport in an aperiodic polariton monotile	Valtýr Kári Daníelsson; Helgi Sigurðsson	2026	arXiv (Cornell University)	0	(link)	
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Floquet Harper-Hofstadter Butterflies and non-Hermitian phase transition in quasicrystals	Mark Kremer; Sebastian Weidemann; Stefano Longhi et al.	2021	Conference on Lasers and Electro-Optics	0	10.1364/cleo_qels.2021.fw2n.2	0.2 (2023)
Advances in classical and quantum wave dynamics on quasiperiodic lattices : a dissertation submitted for the degree of Doctor of Philosophy in Physics, Centre for Theoretical Chemistry and Physics, New Zealand Institute for Advanced Study, Massey University, Albany, New Zealand	Carlo Danieli	2016	Massey Research Online (Massey University)	0	(link)	
Exploring Light-Induced Phases of 2D Materials in a Modulated 1D Quasicrystal	Yifei Bai; Anna R. Dardia; Toshihiko Shimasaki et al.	2026	Physical Review X	0	10.1103/37vk-qz71	
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Multicomponent Photonic Crystals with Inhomogeneous Scatterers	Alexander B. Khanikaev; Mikhail V. Rybin; M. F. Limonov	2012	Series in optics and optoelectronics	0	10.1201/b12175-11	
Time-resolved light propoagation at the band-edge states of 1D Fibonacci quasicrystals	Luca Dal Negro; Claudio J. Otón; Z. Gaburro et al.	2002	MRS Proceedings	0	10.1557/proc-722-l6.9	
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Strong light confinement of tunable resonances in low symmetric quasicrystal through orientational variations	Döne Yılmaz; Mediha Tutgun; Hamza Kurt	2018		0	10.1117/12.2307331	
Topological States in 1D Photonic Quasi-crystals	M.S. Vasconcelos; E.L. Albuquerque	2014	MRS Proceedings	0	10.1557/opl.2014.839	
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Quasicrystalline Structures for Electromagnetic Wave Manipulation	Mikhail V. Rybin	2024		0	10.1109/piers62282.2024.10618664	
		2023	Chinese Optics Letters	0	10.3788/col/2023/21	4.054 (2026)
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Critical analysis of the reentrant localization transition in a one-dimensional dimerized quasiperiodic lattice	Shilpi Roy; Sourav Chattopadhyay; Tapan Mishra et al.	2021	arXiv preprint	0	10.1103/PhysRevB.105.214203	
Localization and topological transitions in non-Hermitian quasiperiodic lattices	Ling-Zhi Tang; Guo-Qing Zhang; Ling-Feng Zhang et al.	2021	Phys. Rev. A 103, 033325 (2021)	0	10.1103/PhysRevA.103.033325	
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Localization of Electronic Wave Functions on Quasiperiodic Lattices	Thomas Rieth; Uwe Grimm; Michael Schreiber	1998	Proceedings of the 6th International Conference on Quasicrystals, Eds. S. Takeuchi and T. Fujiwara (World Scientific, Singapore, 1998), pp. 639-642	0	10.1088/0953-8984/10/4/008	
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Nonmonotonic crossover and scaling behaviors in a disordered 1D quasicrystal	Anuradha Jagannathan; Piyush Jeena; Marco Tarzia	2018	Phys. Rev. B 99, 054203 (2019)	0	10.1103/PhysRevB.99.054203	
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Perpendicular space accounting of localized states in a quasicrystal	Murod Mirzhalilov; M. Ö. Oktel	2020	Phys. Rev. B 102, 064213 (2020)	0	10.1103/PhysRevB.102.064213	
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Emergent multi-loop nested point gap in a non-Hermitian quasiperiodic lattice	Yi-Qi Zheng; Shan-Zhong Li; Zhi Li	2024	Phys. Rev. B 111, 104204 (2025)	0	10.1103/PhysRevB.111.104204	
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Fractal Surface States in Three-Dimensional Topological Quasicrystals	Zhu-Guang Chen; Cunzhong Lou; Kaige Hu et al.	2024	Phys. Rev. Research 6, 043054 (2024)	0	10.1103/PhysRevResearch.6.043054	
Host-atom-driven transformation of a honeycomb oxide into a dodecagonal quasicrystal	Martin Haller; Julia Hewelt; V. Y. M. Rajesh Chirala et al.	2025	Phys. Rev. Lett. 136, 156201 (2026)	0	10.1103/ws7j-tvty	
Closing of gaps and gap labeling and passage from molecular states to critical states in a 2D quasicrystal	Anuradha Jagannathan	2023	arXiv preprint	0	10.1103/PhysRevB.108.115109	
Ground states of a family of frustrated spin models for quasicrystals and their approximants	Anuradha Jagannathan	2025	arXiv preprint	0	10.1103/ht6g-5dwl	
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Density Distribution in Soft Matter Crystals and Quasicrystals	Priya Subramanian; Daniel J. Ratliff; Alastair M. Rucklidge et al.	2020	Phys. Rev. Lett. 126, 218003 (2021)	0	10.1103/PhysRevLett.126.218003	
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Scale-free localization versus Anderson localization in unidirectional quasiperiodic lattices	Yu Zhang; Luhong Su; Shu Chen	2025	Phys. Rev. B 111, L140201(2025)	0	10.1103/PhysRevB.111.L140201	
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Fractalized magnon transport on the quasicrystal with enhanced stability	Junmo Jeon; Se Kwon Kim; SungBin Lee	2022	arXiv preprint	0	10.1103/PhysRevB.106.134431	
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High-harmonic spectroscopy of mobility edges in one-dimensional quasicrystals	H. K. Avetissian; B. R. Avchyan; A. Brown et al.	2025	arXiv preprint	0	(link)	
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Tensor network method for real-space topology in quasicrystal Chern mosaics	Tiago V. C. Antão; Yitao Sun; Adolfo O. Fumega et al.	2025	Phys. Rev. Lett. 136, 156601 (2026)	0	10.1103/hhdf-xpwg	
Interlayer Decoupling in 30° Twisted Bilayer Graphene Quasicrystal	Bing Deng; Binbin Wang; Ning Li et al.	2020	ACS Nano (2020)	0	10.1021/acsnano.9b07091	17.894 (2026)
Spin waves in one-dimensional bi-component quasicrystals	J. Rychły; J. W. Kłos; M. Mruczkiewicz et al.	2014	Phys. Rev. B 92, 054414 (2015)	0	10.1103/PhysRevB.92.054414	
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A parametric study of the lensing properties of dodecagonal photonic quasicrystals	E. Di Gennaro; D. Morello; C. Miletto et al.	2007	arXiv preprint	0	10.1016/j.photonics.2007.12.001	
Exact anomalous mobility edges in one-dimensional non-Hermitian quasicrystals	Xiang-Ping Jiang; Weilei Zeng; Yayun Hu et al.	2024	arXiv preprint	0	(link)	
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Relationship between Structure and Dynamics of an Icosahedral Quasicrystal using Unsupervised Machine Learning	Edwin A. Bedolla-Montiel; Susana Marín-Aguilar; Marjolein Dijkstra	2025	arXiv preprint	0	(link)	
Quantum bridge states and sub-dimensional transports in quasicrystals	Junmo Jeon; SungBin Lee	2022	arXiv preprint	0	(link)	
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Localization transition in weakly-interacting Bose superfluids in one-dimensional quasiperdiodic lattices	Samuel Lellouch; Laurent Sanchez-Palencia	2014	Physical Review A, American Physical Society, 2014, 90, pp.061602(R)	0	10.1103/PhysRevA.90.061602	
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Gapless superconductivity and its real-space topology in quasicrystals	Kazuma Saito; Masahiro Hori; Ryo Okugawa et al.	2024	Physical Review Research 7, L022077 (2025)	0	10.1103/nb7l-f44r	
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Non-diffusion transport in decoherent non-Hermitian quasicrystals	Yudong Ren; Rui Zhao; Kangpeng Ye et al.	2025	arXiv preprint	0	(link)	
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Space-Time Quasicrystal Structures and Inflationary and Late Time Evolution Dynamics in Accelerating Cosmology	Sergiu I. Vacaru	2018	Clas. Quant. Grav. 35 (2018) no. 24, 245009	0	10.1088/1361-6382/aaec93	
Quasiperiodic pairing in graphene quasicrystals	Rasoul Ghadimi; Bohm-Jung Yang	2024	Nano Lett. 2025, 25, 5, 1808 1815	0	10.1021/acs.nanolett.4c04386	
Electronic states of a disordered 2D quasiperiodic tiling: from critical states to Anderson localization	Anuradha Jagannathan; Marco Tarzia	2022	arXiv preprint	0	10.1103/PhysRevB.107.054206	
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Starobinsky Inflation and Dark Energy and Dark Matter Effects from Quasicrystal Like Spacetime Structures	R. Aschheim; L. Bubuianu; Fang Fang et al.	2016	Annals of Physics 394 (2018) 120-138	0	10.1016/j.aop.2018.04.033	
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Localization Transitions in a Half-Filled Helical Aubry-André Model	Taylan Yildiz; B. Tanatar; Balázs Hetényi	2026	arXiv preprint	0	10.1103/c5wk-3ym5	
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The scales of disorder in perfect quasicrystals	Alan Rodrigo Mendoza Sosa; Atahualpa S. Kraemer; Michael Schmiedeberg et al.	2026	arXiv preprint	0	(link)	
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Unveiling exotic magnetic phase diagram of a non-Heisenberg quasicrystal approximant	Farid Labib; Kazuhiro Nawa; Shintaro Suzuki et al.	2023	arXiv preprint	0	(link)	
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Striking Similarities in Dynamics and Vibrations of 2D Quasicrystals and Supercooled Liquids	Edwin A. Bedolla-Montiel; Marjolein Dijkstra	2025	arXiv preprint	0	(link)	
Cholesteric shells: two-dimensional blue fog and finite quasicrystals	Livio Nicola Carenza; Giuseppe Gonnella; Davide Marenduzzo et al.	2021	Physical Review Letters 2022	0	10.1103/PhysRevLett.128.027801	
Mapping of strained graphene into one-dimensional Hamiltonians: quasicrystals and modulated crystals	Gerardo G. Naumis; Pedro Roman-Taboada	2014	arXiv preprint	0	10.1103/PhysRevB.89.241404	
The fate of Wannier-Stark localization and skin effect in periodically driven non-Hermitian quasiperiodic lattices	Aditi Chakrabarty; Sanjoy Datta	2024	arXiv preprint	0	(link)	
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Quasilattice-conserved optimization of the atomic structure of decagonal Al-Co-Ni quasicrystals	Xiao-Tian Li; Xiao-Bao Yang; Yu-Jun Zhao	2014	arXiv preprint	0	10.1088/0256-307X/32/3/036102	
Quasiperiodic Skin Criticality in an Exactly Solvable Non-Hermitian Quasicrystal	Zhangyuan Chen; Muhammad Idrees; Ying Yang et al.	2026	Phys. Rev. B 113, 235423(2026)	0	10.1103/wlsw-zyq9	
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Symmetry-breaking dynamics of the finite-size Lipkin-Meshkov-Glick model near ground state	Yi Huang; Tongcang Li; Zhang-qi Yin	2017	Phys. Rev. A 97, 012115 (2018)	0	10.1103/PhysRevA.97.012115	
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A general approach to the exact localized transition points of 1D mosaic disorder models	Yanxia Liu	2022	arXiv preprint	0	(link)	
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Self-duality of One-dimensional Quasicrystals with Spin-Orbit Interaction	Deepak Kumar Sahu; Aruna Prasad Acharya; Debajyoti Choudhuri et al.	2021	Phys. Rev. B 104, 054202 (2021)	0	10.1103/PhysRevB.104.054202	
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Strictly Localized States on the Socolar Dodecagonal Lattice	M. Akif Keskiner; M. Ö Oktel	2022	Phys. Rev. B 106, 064207 (2022)	0	(link)	
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Interplay of Quasiperiodic Criticality and the Non-Hermitian Skin Effect	Zhangyuan Chen; Xianqi Tong; Xiaosen Yang	2026	arXiv preprint	0	(link)	
Computing with almost periodic functions	R. V. Moody; M. Nesterenko; J. Patera	2008	arXiv preprint	0	(link)	
Quantum criticality and Kibble-Zurek scaling in the Aubry-André-Stark model	En-Wen Liang; Ling-Zhi Tang; Dan-Wei Zhang	2024	Phys. Rev. B 110, 024207 (2024)	0	10.1103/PhysRevB.110.024207	
Single-particle and many-body analyses of a quasiperiodic integrable system after a quench	Kai He; Lea F. Santos; Tod M. Wright et al.	2013	Phys. Rev. A 87, 063637 (2013)	0	10.1103/PhysRevA.87.063637	
Coexistence of topological Anderson insulator and multifractal critical phase in a non-Hermitian quasicrystal	Qi-Bo Zeng; Rong Lü	2026	Phys. Rev. B 113, 224203 (2026)	0	10.1103/n9rf-zk9t	
Quasicrystalline Spin Liquid	Sunghoon Kim; Mohammad Saad; Dan Mao et al.	2024	Phys. Rev. B 110, 214438, 2024	0	10.1103/PhysRevB.110.214438	
Disordered, Quasicrystalline and Crystalline Phases of Densely Packed Tetrahedra	Amir Haji-Akbari; Michael Engel; Aaron S. Keys et al.	2010	Nature 462, 773–777 (2009)	0	10.1038/nature08641	
Photoinduced Phase Transition in Two-Band model on Penrose Tiling	Ken Inayoshi; Yuta Murakami; Akihisa Koga	2021	J. Phys.: Conf. Ser. 2164, 012050 (2022)	0	10.1088/1742-6596/2164/1/012050	
Quasicrystalline electronic states in 30° rotated twisted bilayer graphene	Pilkyung Moon; Mikito Koshino; Young-Woo Son	2019	Phys. Rev. B 99, 165430 (2019)	0	10.1103/PhysRevB.99.165430	
Observation of subdiffusion of a disordered interacting system	E. Lucioni; B. Deissler; L. Tanzi et al.	2010	Phys. Rev. Lett. 106, 230403 (2011)	0	10.1103/PhysRevLett.106.230403	
Correlation function of weakly interacting bosons in a disordered lattice	B. Deissler; E. Lucioni; M. Modugno et al.	2010	New J. Phys. 13 (2011) 023020	0	10.1088/1367-2630/13/2/023020	
Theory of spin Bott index for quantum spin Hall states in non-periodic systems	Huaqing Huang; Feng Liu	2018	Phys. Rev. B 98, 125130 (2018)	0	10.1103/PhysRevB.98.125130	
Time-quasiperiodic topological superconductors with Majorana Multiplexing	Yang Peng; Gil Refael	2018	Phys. Rev. B 98, 220509 (2018)	0	10.1103/PhysRevB.98.220509	
Hubbard Models for Quasicrystalline Potentials	Emmanuel Gottlob; Ulrich Schneider	2022	Phys. Rev. B 107, 144202 (2023)	0	10.1103/PhysRevB.107.144202	
Affine Dihedral Subgroups of Higher Dimensional Cubic Lattices \$\mathbb{Z}^n\$ and Quasicrystallography	Nazife Ozdes Koca; Mehmet Koca; Ramazan Koc et al.	2023	Sultan Qaboos University Journal For Science 2025 : Vol. 30: Iss. 2, 102-116	0	10.53539/2414-536X.1405	
Magnetic states induced by electron-electron interactions in a plane quasiperiodic tiling	A. Jagannathan; H. J. Schulz	1996	arXiv preprint	0	10.1103/PhysRevB.55.8045	
Amplitude-dependent edge states and discrete breathers in nonlinear modulated phononic lattices	Matheus I. N. Rosa; Michael J. Leamy; Massimo Ruzzene	2022	arXiv preprint	0	(link)	
Superconducting properties of Fibonacci chains with enhanced superconducting pairing at the boundaries	Quanyong Zhu; Guo-Qiao Zha; A. A. Shanenko et al.	2024	Physical Review B 112, 134503 (2025)	0	10.1103/j8tj-82ty	
Chemical localization	V. F. Gantmakher	2002	UFN 172(11), 1283 (2002) [English translation in Physics-Uspekhi 45(11), (2002)]	0	10.1070/PU2002v045n11ABEH001246	
Sub-diffusive electronic states in octagonal tiling	G. Trambly de Laissardière; C. Oguey; D. Mayou	2016	Journal of Physics: Conf. Series 809 (2017) 012020	0	10.1088/1742-6596/809/1/012020	
Phase transitions and bunching of correlated particles in a non-Hermitian quasicrystal	Stefano Longhi	2023	Phys. Rev. B 108, 075121 (2023)	0	10.1103/PhysRevB.108.075121	
Higher-dimensional Hofstadter butterfly on Penrose lattice	Rasoul Ghadimi; Takanori Sugimoto; Takami Tohyama	2022	arXiv preprint	0	10.1103/PhysRevB.106.L201113	
Lyapunov exponent for pure and random Fibonacci chains	M. T. Velhinho; I. R. Pimentel	2000	Phys. Rev. B 61, 1043 (2000)	0	10.1103/PhysRevB.61.1043	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Exploring Aperiodic Order in Photonic Time Crystals	Marino Coppolaro; Massimo Moccia; Giuseppe Castaldi et al.	2025	arXiv preprint	0	(link)	
Ground state of a two dimensional quasiperiodic antiferromagnet	A. Jagannathan	2004	arXiv preprint	0	10.1103/PhysRevB.71.115101	
On the Spectra of Sieved Schrödinger Operators	Jake Fillman; Alexandro Luna	2025	arXiv preprint	0	(link)	
Complete Structural Determination of Mesostructural Dodecagonal Quasicrystalline Particles	Xueliang Zhang; Xi Wang; Nobuhisa Fujita et al.	2026	arXiv preprint	0	(link)	
Antiferromagnetic order in the Hubbard Model on the Penrose Lattice	Akihisa Koga; Hirokazu Tsunetsugu	2017	Phys. Rev. B 96, 214402 (2017)	0	10.1103/PhysRevB.96.214402	
Mott transition for a Lieb-Liniger gas in a shallow quasiperiodic potential: Delocalization induced by disorder	Hepeng Yao; Luca Tanzi; Laurent Sanchez-Palencia et al.	2024	Phys. Rev. Lett. 133, 123401 (2024)	0	10.1103/PhysRevLett.133.123401	
Understanding shape entropy through local dense packing	Greg van Anders; Daphne Klotsa; N. Khalid Ahmed et al.	2013	PNAS 111, E4812-E4821 (2014)	0	10.1073/pnas.1418159111	
Approximating Metal-Insulator Transitions	C. Danieli; K. Rayanov; B. Pavlov et al.	2014	arXiv preprint	0	10.1142/S0217979215500368	
Engineering bands of extended electronic states in a class of topologically disordered and quasiperiodic lattices	Biplab Pal; Arunava Chakrabarti	2014	Physics Letters A, Volume 378, Issue 37, Page 2782 (2014)	0	10.1016/j.physleta.2014.07.034	
Combined energy – diffraction data refinement of decagonal AlNiCo	M. Mihalkovič; C. L. Henley; M. Widom	2003	arXiv preprint	0	10.1016/j.jnoncrysol.2003.11.034	
Electronic and optical properties of strained graphene and other strained 2D materials: a review	Gerardo G. Naumis; Salvador Barraza-Lopez; Maurice Oliva-Leyva et al.	2016	Rep. Prog. Phys. 80 096501 (2017)	0	10.1088/1361-6633/aa74ef	
Fate of Majorana zero modes by a modified real-space-Pfaffian method and mobility edges in a one-dimensional quasiperiodic lattice	Shujie Cheng; Yufei Zhu; Gao Xianlong et al.	2021	arXiv preprint	0	(link)	
Stochastic Flips on Dimer Tilings	Thomas Fernique; Damien Regnault	2011	arXiv preprint	0	(link)	
Cleaved surface of i-AlPdMn quasicrystals: Influence of the local temperature elevation at the crack tip on the fracture surface roughness	Laurent Ponson; Daniel Bonamy; Luc Barbier	2006	Physical Review B 74 (20/11/2006) 184205	0	10.1103/PhysRevB.74.184205	
Resonant metallic states in driven quasiperiodic lattices	L. Morales-Molina; E. Doerner; C. Danieli et al.	2014	arXiv preprint	0	(link)	
On the spectrum of 1D quantum Ising quasicrystal	W. N. Yessen	2011	arXiv preprint	0	(link)	
Emergent periodic and quasiperiodic lattices on surfaces of synthetic Hall tori and synthetic Hall cylinders	Yangqian Yan; Shao-Liang Zhang; Sayan Choudhury et al.	2018	Phys. Rev. Lett. 123, 260405 (2019)	0	10.1103/PhysRevLett.123.260405	
Observation of an aperiodic polariton monotile	Sergey Alyatkin; Yaroslav V. Kartashov; Kirill Sitnik et al.	2026	arXiv preprint	0	(link)	
Unraveling Multifractality and Mobility Edges in Quasiperiodic Aubry-André-Harper Chains through High-Harmonic Generation	Marlena Dziurawiec; Jessica O. de Almeida; Mohit Lal Bera et al.	2023	arXiv preprint	0	10.1103/PhysRevB.110.014209	
Phase Diagram of Hard Tetrahedra	Amir Haji-Akbari; Michael Engel; Sharon C. Glotzer	2011	J. Chem. Phys. 135, 194101 (2011)	0	10.1063/1.3651370	
Superconductivity on a Quasiperiodic Lattice: Extended-to-Localized Crossover of Cooper Pairs	Shiro Sakai; Nayuta Takemori; Akihisa Koga et al.	2016	Phys. Rev. B 95, 024509 (2017)	0	10.1103/PhysRevB.95.024509	
Subdiffusion of nonlinear waves in quasiperiodic potentials	Marco Larcher; Tetyana V. Laptyeva; Joshua D. Bodyfelt et al.	2012	New Journal of Physics 14 (2012) 103036	0	10.1088/1367-2630/14/10/103036	
Diffraction and pseudospectra in non-Hermitian quasiperiodic lattices	Ananya Ghatak; Dimitrios H. Kaltsas; Manas Kulkarni et al.	2024	arXiv preprint	0	(link)	
Discrete quasiperiodic sets with predefined covering cluster	Nicolae Cotfas	2005	Philosophical Magazine 86 (2006) 895-900	0	10.1080/14786430500263413	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Localization control born of intertwined quasiperiodicity and non-Hermiticity	Jeon, Junmo; Lee, Sung-Bin	2022	SciPost Phys.Core	0	10.21468/SciPostPhysCore.6.4.077	
Topological Pumping over a Photonic Fibonacci Quasicrystal	Verbin, Mor; Zilberberg, Oded; Lahini, Yoav et al.	2014	Phys.Rev.B	0	10.1103/PhysRevB.91.064201	
Anderson localization in Bose-Einstein condensates	Modugno, Giovanni	2010	Rept.Prog.Phys.	0	10.1088/0034-4885/73/10/102401	
Support Vector Machine Kernels as Quantum Propagators	Kuo, Nan-Hong; Wong, Renata	2025	INSPIRE record	0	(link)	
Anomalous localization in a kicked quasicrystal	Shimasaki, Toshihiko; Prichard, Max; Kondakci, H. Esat et al.	2022	Nature Phys.	0	10.1038/s41567-023-02329-4	
Colloquium: Synthetic quantum matter in non-standard geometries	Grass, Tobias; Bercioux, Dario; Bhattacharya, Utso et al.	2024	Rev.Mod.Phys.	0	10.1103/RevModPhys.97.011001	
Observing the two-dimensional Bose glass in an optical quasicrystal	Yu, Jr-Chiun; Bhave, Shaurya; Reeve, Lee et al.	2023	Nature	0	10.1038/s41586-024-07875-2	
Phasonic Spectroscopy of a Quantum Gas in a Quasicrystalline Lattice	Rajagopal, Shankari V.; Shimasaki, Toshihiko; Dotti, Peter et al.	2019	Phys.Rev.Lett.	0	10.1103/PhysRevLett.123.223201	
Non-Hermitian physics	Ashida, Yuto; Gong, Zongping; Ueda, Masahito	2020	Adv.Phys.	0	10.1080/00018732.2021.1876991	
Cavity QED with Quantum Gases: New Paradigms in Many-Body Physics	Mivehvar, Farokh; Piazza, Francesco; Donner, Tobias et al.	2021	Adv.Phys.	0	10.1080/00018732.2021.1969727	

Source overlap

Source	Retrieved	Found only here	Filtered venues
openalex	656	599	37
arxiv	613	567	0
inspire	17	10	0

'Found only here' counts unique records no other source returned – a measure of each database's marginal contribution.

Distributions

Publication year

Year	Records
1994	1
1996	1
1997	1
1998	2
1999	3
2000	2
2001	3
2002	5
2003	3
2004	5
2005	4
2006	6
2007	9
2008	7
2009	8
2010	10
2011	9
2012	9
2013	9
2014	23
2015	25
2016	64
2017	56
2018	84
2019	99
2020	89
2021	79
2022	87
2023	94
2024	152

Year	Records
2025	154
2026	123

Top venues

Venue	Records	OpenAlex 2-yr mean citedness
arXiv preprint	354	
arXiv (Cornell University)	57	
Advanced Energy Materials	34	18.63 (2026)
Advanced Materials	23	22.086 (2026)
Advanced Functional Materials	19	14.975 (2026)
ACS Applied Materials & Interfaces	19	8.382 (2026)
The Journal of Physical Chemistry C	16	
Nature Energy	15	
ACS Energy Letters	15	16.712 (2026)
The Journal of Chemical Physics	13	
Journal of the American Chemical Society	12	
Nano Energy	12	
Journal of Materials Chemistry A	11	
Optics Express	11	
Science	10	

Most frequent authors

Author	Records
Michaël Grätzel	21
Marco Bernasconi	19
Shaik M. Zakeeruddin	15
Zheyong Fan	14
Nam-Gyu Park	13
Omar Abou El Kheir	13
Shu Chen	11
Volker L. Deringer	10
Stefano Longhi	10
Un-Gi Jong	10
Chol-Jun Yu	10
Longwen Zhou	10
Anders Hagfeldt	9
Michael D. McGehee	9
Henry J. Snaith	9

Journal metrics

Metric: OpenAlex 2-yr mean citedness (openalex_2yr), from lit/journals/metrics.json (journals.py). Values are kept per year; the evolution table shows every year on file for venues that appear in this record set.

Venues in this set by OpenAlex 2-yr mean citedness

Venue	Records	OpenAlex 2-yr mean citedness	Year	Q
Chemical Reviews	1	53.886	2026	
Chemical Society Reviews	1	42.455	2026	
Energy & Environmental Science	1	30.475	2026	
Advanced Materials	23	22.086	2026	
Advanced Energy Materials	34	18.63	2026	
ACS Nano	3	16.894	2026	
ACS Energy Letters	15	16.712	2026	
Angewandte Chemie International Edition	6	15.006	2026	
Advanced Functional Materials	19	14.975	2026	
Energy Storage Materials	1	13.488	2026	
Cement and Concrete Research	1	13.05	2026	
Chem	2	11.7	2026	
Applied Physics Reviews	1	11.309	2026	
Chemical Engineering Journal	3	10.953	2026	
Acta Materialia	3	10.171	2026	
ACS Materials Letters	1	9.392	2026	
Artificial Intelligence Chemistry	1	9.034	2026	
Advanced Science	6	8.903	2026	
Current Opinion in Chemical Engineering	1	8.446	2026	
ACS Applied Materials & Interfaces	19	8.382	2026	
EcoMat	1	7.557	2026	
Chemistry of Materials	3	6.387	2026	
Cell Reports Physical Science	1	5.933	2026	
Chaos Solitons & Fractals	1	5.898	2026	
Ceramics International	1	5.869	2026	

Filtered non-curated venues

Venue	Records removed
Figshare	15
Zenodo	11
SSRN Electronic Journal	9
Open MIND	1
Preprints.org	1

Count history

Per-block totals across archived runs (counts_history.csv); drift shows how the indexes – or your queries – changed.

Block	20260815T095908	20260828T091142	20260828T100120	20260828T160426
PSC	-	-	14,620	4,744
MLP	-	239	838	376
NOV	-	0	0	0
QC	-	-	978	585

Suggestions

- Block PSC: openalex 4,569 hits – a generic term is probably driving this; tighten a group or add a more specific one before reading.
- Block PSC: counts differ >20x across backends (175 to 4,569); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- Block NOV: zero hits on every backend. Either the intersection is genuinely empty (a finding – check the synonyms first) or one group is too narrow; try dropping one group and rerunning.
- Block QC: counts differ >20x across backends (17 to 414); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- 2 block/backend pair(s) hit the `--limit` cap (300); raise it (largest total 4,569) if you need the complete record set rather than the most-cited slice.
- No citation-grade backend (Scopus, Web of Science, NASA ADS) was in this search; add ADS (free token) or Scopus before quoting counts.
- Open-access status was not looked up; rerun with `--pdfs` (optionally `--pdf-blocks`) to collect legal OA PDF links via Unpaywall.
- The PRISMA flow's manual stages are empty: fill in prisma.json (screened / excluded / assessed / included) and rerun report.py to complete the diagram.
- Block PSC: total hits changed from 14,620 (run 20260828T100120) to 4,744; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.
- Block MLP: total hits changed from 838 (run 20260828T100120) to 376; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.
- Block QC: total hits changed from 978 (run 20260828T100120) to 585; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.